Appendix A

Measuring the Economic Impacts of U.S. Tariffs on Canada: A National, Regional and Industrial Analysis

By

Richard J. DiFrancesco, Ph.D., M.C.I.P., R.P.P.

Table of Contents

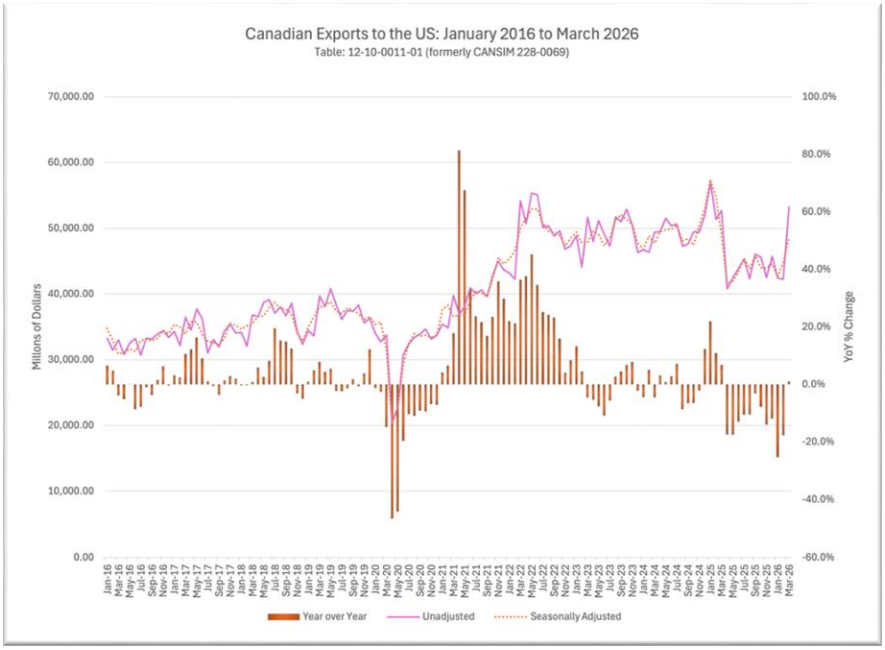
MEASURING THE ECONOMIC IMPACTS OF U.S. TARIFFS ON CANADA: A NATIONAL, REGIONAL AND INDUSTRIAL ANALYSIS 1

1. U.S. Tariffs and Canada's Economy 3
2. Tariff Impact Scenarios 7
2.1 Reduced Household Consumption Expenditures 7
2.2 Agriculture, Agri-food and Seafood Exports to the U.S. 8
2.4 Declining Steel and Aluminum Exports to the U.S. 10
2.5 Softwood Lumber/Forestry Exports to the U.S. 11
2.6 Automobile and Automotive Parts Exports to the U.S. 12
3.0 Measuring the Economic Impacts of Tariff Scenarios 12
3.1 National Impacts by Industry – 'Open Model' Results 14
3.2 National Impacts by Industry – 'Closed Model' Results 26
3.3 Provincial Distribution of Gross Output/Value-Added Impacts 29
3.4 The Varying Nature of Economic Impacts Across Regions in Canada 32
3.4 Provincial Distribution of Employment Impacts 35
5.0 Discussion 37

1. U.S. Tariffs and Canada's Economy

Figure 1 shows that Canada's trade relationship with the United States (U.S.) has been turbulent for some time, though prior to 2025 much of this was driven by facttors like global recessions and pandemics. The most recent period in our trading history with the U.S. exhibits highly chaotic behaviour driven largely by a radical shift in U.S. policy regarding international trade with Canada as represented in a spate of new tariffs imposed by President Trump early in 2025. Indeed, this new tariff regime has been referred to by Canadian Prime Minister Carney, as a 'rupture' in the rules-based system of governance that has presided over western trade for much of the post-War era. What follows is an examination of the impacts associated with various tariff scenarios across industries and regions in Canada.

Figure 1:[1]



As of October 2025, Canada's domestic exports to the U.S. accounted for more than 67 percent of Canada's total exports by value – the lowest level (barring pandemic-related distortions) since the current methodology of computing trade balances was established in 1997 (Ljuggren, 2026).[2] Statistics Canada (2025) notes that in 2024, some 53,448

[1] Source: author's calculations and Statistics Canada Table: 12-10-0011-01 (formerly CANSIM 228-0069).

[2] Ljunggren, D. (2026). "Canada''s share of exports to U.S. hit lowest level in decades in October." CBC News https://www.cbc.ca/news/business/canada-trade-oct-2025-9.7037913 – Posted January 8, 2026

establishments in Canada exported goods abroad and that nearly 87 percent of these exports were directed toward the U.S.. And, according to the results of the Canadian Survey on Business Conditions for Q3 2025, those businesses which exported goods or services to the U.S. over the past 12 months made an average of 36.8 percent of their total sales in that market, and one-third of these businesses reported having tariffs imposed on them by the U.S.. This share rose to nearly 65 percent for firms involved in wholesale trade, and more than half (53.2 percent) for firms in manufacturing. More than 55 percent of those firms reported that these tariffs would have a negative impact on their operations. The report also notes that "[o]ver the next three months, nearly two-fifths of businesses exporting to the United States expect their operating expenses to increase (39.4 percent) or foresee a decline in their profitability (39.3 percent), and over one-quarter (25.7 percent) expect their overall exports to decrease."[3] Three-quarters of responding exporting firms indicated that they planned to take actions to address these expectations over the next 12 months including seeking alternative markets (24.2 percent) and delaying major investments or expenditures (16.5 percent). Of those responding firms active in manufacturing, more than 90 percent indicated their intention to take actions over the next year, and more than one-third (33.6 percent) indicated that they will be seeking alternative customers outside of the U.S..[4] Clearly, the long-established 'stable and predictable' relationship which has characterized North American trade for the better part of one century has changed dramatically. Given that the economies of Canada and the U.S. are so deeply inter-woven, these shocks portend significant economic impacts on both sides of the border. This essay represents an attempt to develop some clearer pictures of how these impacts could manifest under a variety of tariff scenarios.

Most of the advisory reports and updates coming out of major Canadian banks, federal and provincial agencies, and even some quasi-governmental and private research groups tend to agree that a scenario where the existing Canada-United States-Mexico Trade Agreement (or CUSMA) is completely terminated and replaced with a series of crippling sector-specific and blanket tariffs as unlikely. Indeed, Scotiabank Economics (2026a) refers to such an eventuality as a 'tail event'.[5] Though, it is worthwhile noting that the Bank of Canada's degree of optimism regarding the ongoing resilience of Canada-U.S. trade (and specifically

---

[3] Statistics Canada (2025). "Impact of tariffs on businesses in Canada: Expectations and strategic responses, third quarter of 2025." (https://www150.statcan.gc.ca/n1/en/pub/11-621-m/11-621-m2025012-eng.pdf?st=8b4HwfK_) published on September 18, 2025.

[4] Statistics Canada (2025). "Impact of tariffs on businesses in Canada: Expectations and strategic responses, third quarter of 2025." (https://www150.statcan.gc.ca/n1/en/pub/11-621-m/11-621-m2025012-eng.pdf?st=8b4HwfK_) published on September 18, 2025.

[5] Scotiabank Economics (2026a). "Low Probability, High Cost: The Macroeconomics of Abandoning CUSMA." (https://www.scotiabank.com/ca/en/about/economics/economics-publications/post.other-publications.insights-views.cusma-scenario-note--march-5--2026-.html)

of Canadian domestic exports to the U.S.) seems to be steadily eroding with each passing quarter. While unlikely, Scotiabank Economics (2026a) does model the repercussions of one such scenario (their "severe fragmentation" scenario) and the results show that such an eventuality would likely push the Canadian economy into a recession (as well as causing significant hardship in the U.S. economy).[6]

Shocks associated with the failure of CUSMA (or even a significant weakening of its current protections) would be transmitted to U.S. and Canadian economies via four primary vectors (or 'channels') including:

1. **Lowering of trade volumes**: failure of CUSMA would make all Canadian exports less competitive to U.S. customers virtually overnight which could cause exports to the U.S. from Canada to decline. The aforementioned 2026 report by Scotiabank Economics[7] estimated that Canadian exports to the U.S. could decline by anywhere from 1 to 4 percent in very short order. Interestingly, there is plenty of 'positive' economic news regarding the Canadian economy and its apparent resilience in the face of this new tariff regime. As discussed in a 2026 report by RBC Economics, much of this apparent buoyancy is result of price-driven export surges of precious minerals and energy, and not an indication of resilience on behalf of those sectors directly impacted by tariffs.[8]
2. **Weakening productivity**: Scotiabank Economics (2026a) also notes that the dismantling of long-established North American supply chains would push both countries into less efficient production structures as "…capital is stranded, labour is

[6] Scotiabank Economics (2026a). "Low Probability, High Cost: The Macroeconomics of Abandoning CUSMA." (https://www.scotiabank.com/ca/en/about/economics/economics-publications/post.other-publications.insights-views.cusma-scenario-note--march-5--2026-.html)

[7] Scotiabank Economics (2026a). "Low Probability, High Cost: The Macroeconomics of Abandoning CUSMA." (https://www.scotiabank.com/ca/en/about/economics/economics-publications/post.other-publications.insights-views.cusma-scenario-note--march-5--2026-.html)

[8] RBC Economics (May 20, 2026) notes that the diversification of Canada's export portfolio has become a key policy concern for the Federal Government (and indeed lower governments). Results over Q4 2024 to Q4 2025 suggest that the year-over-year decline in nominal merchandise exports declined 0.8 % or $5.9 billion y-o-y which is far less than that was originally feared and has led some to conclude that the Canadian economy is exhibiting resilience in the face of tariffs. This RBC note however states that much of this apparent resilience is derived primarily from a price induced surge in gold exports as well as the effects of new pipeline capacity which came online recently thereby increasing oil and gas exports. This 'resilience' is not the result, therefore, of the diversification of Canada's exports in highly tariffed sectors like steel products, automobiles and parts, and wood products (i.e., these key sectors have not found alternative markets). https://www.rbc.com/en/economics/canadian-analysis/featured-analysis/insights/surging-gold-prices-inroads-to-foreign-markets-cushion-canadas-exports/

underutilized, and firms are pushed toward higher cost domestic substitutes, hence leading to higher prices paid by businesses and households."[9]

3. **Declining confidence on behalf of households and firms:** where escalating uncertainty regarding the rules of the game is negatively impacting the confidence of consumers (i.e., fear of job losses) and businesses (i.e., reconsidering domestic or trans-border investments); and,
4. **Rising costs due to Inflation passthrough:** where tariffs result in increased prices both directly and indirectly which in turn leads to higher prices for both households and firms.

Global Affairs Canada (2025) in its Quarterly Economic and Trade Report for the second quarter of 2025 (Q2) note that "Canada's exports of goods and services declined by 10.6 percent in Q2…" with decreases observed across most goods and service categories and that this drop was driven largely by a nearly 19 percent decline in exports to the U.S. in Q2.[10] Interestingly, Global Affairs Canada (2025a) also notes that Canadian exports to non-U.S. destinations surged by nearly 15 percent in Q2 which partially offset the losses attributable to U.S. tariffs, especially Section 232 tariffs (but again, much of this has been driven by price-induced surges in demand for Canadian precious minerals and energy resources).

Should Scotiabank Economics' (2026a) 'Severe Fragmentation' scenario materialize – where CUSMA exemptions are lost and Section 232-like sector specific tariffs are maintained or increased (in combination with some form of a blanket tariff) - export volumes could decline by 4 percent relative to current levels. It is worth noting that at the time of writing overall export volumes to the U.S. have dropped by 2 percent (in response to the current tariff regime), so combining this with Scotiabank's forecast could see exports down by 6 percent by one year post-CUSMA failure, and this would translate into a 2 percent drop in Canadian GDP).

Far more likely is a scenario whereby sector-specific (Section 232 or other) tariffs persist in some form and hit targeted industries and regions disproportionately (these outcomes will be discussed below). However, while unlikely, implementation of severe protectionism by the U.S. and the elimination of CUSMA exemptions is a possibility that must be considered. Indeed, in its January 2025 Monetary Policy Report, the Bank of Canada noted that "[e]stimates of the sensitivity of Canadian exports to price changes are based on historical data and are found to be quite low. However, in the case of such a large shock, it is possible

---

[9] Scotiabank Economics (2026a). "Low Probability, High Cost: The Macroeconomics of Abandoning CUSMA." (https://www.scotiabank.com/ca/en/about/economics/economics-publications/post.other-publications.insights-views.cusma-scenario-note--march-5--2026-.html)

[10] Global Affairs Canada (2025). "Quarterly Economic and Trade Report: 2025 – Quarter 2." ISSN 2819-4063 (https://international.canada.ca/en/global-affairs/corporate/reports/chief-economist/quarterly/2025-q2) – published September, 2025.

that exports could fall by considerably more than predicted by the Bank's projection models. Increasing the sensitivity for both commodity and non-commodity exports to U.S. tariffs results in a more significant decline in exports and business investment and, as a result, weaker GDP growth.... It also leads to a more significant depreciation of the Canadian dollar because Canada's trade balance worsens by more than with the benchmark calibration."[11] The Bank's (very conservative) projections call for growth in personal consumption in Canada from 2024 through 2026 of around 1 percent (again, largely associated with positive trajectories for metals and energy resources). They note, however, that substantial tariff-induced declines to these levels are possible.[12]

Two general types of tariff impact scenarios have been proposed, discussed, and analyzed by various private- and public-sector actors since trade tensions between Canada and the U.S. began to increase. The first type typically includes depictions of change in a variety of high-level macroeconomic variables (e.g., declining trade volumes, cessation of private-sector investment born out of increasing uncertainty, and the rapid deceleration of household consumption expenditures in the face of increasing employment precarity, etc.). The second general type is more industry-specific and tends to address those sectors of the Canadian economy which are most vulnerable to Section 232 (and more recently, Section 238) Tariffs imposed by Washington (e.g., on steel, aluminum, soft-wood lumber, automobiles, automotive parts, and agriculture and agri-food) and the regions of Canada that are heavily invested in them. The following sections will discuss a selection of these scenarios, pose a series of reasonable scenarios based on current thinking, and then analyze their potential economic impacts on Canada's industries and regions.

2. Tariff Impact Scenarios

2.1 Reduced Household Consumption Expenditures

Scotiabank Economics (2026a) presents the results of two modeled tariff impact scenarios - a "contained" scenario where GDP drops by 0.6 percent and a "severe fragmentation"

[11] Bank of Canada (2025). "Monetary Policy Report – January, 2025." (https://www.bankofcanada.ca/wp-content/uploads/2025/01/mpr-2025-01-29.pdf)

[12] This caveat by the Bank of Canada is noteworthy. The Bank of Canada is essentially saying that their baseline projections are based on elasticities of substitution (i.e., domestic for imported goods) derived from historical data collected in a relatively stable trade environment. This note suggests that this historical pattern may not serve as a suitable model for what could happen going forward in a period of unprecedented trade uncertainty. In other words, the Bank of Canada's base case scenarios (and associated recommendations) are based on parameters estimated on historical data pertaining to a system that differs qualitatively from what comes next. This is an ever-present concern when relying on time-series analysis where the past is used as a guide for predictions of future behaviour.

scenario (which is basically a CUSMA-failure scenario) where GDP drops by 1.9 percent.[13] In each case, these effects are projected at the one-year mark. In the severe scenario they note that real consumption contracts, especially for durable goods as weaker exports to the U.S. reduce GDP, employment and household income in Canada, thereby depressing consumption. Similarly, the Bank of Canada's 2025 baseline scenario is based on the presumed continuation of CUSMA (in some form) but still envisions Canadian GDP at approximately 1.5 percent below its January 2025 projection by the end of 2026.[14] In the end, it appears as though significant reductions in CUSMA protections (and/or increased sector-specific and blanket tariffs) would probably shave around half a percentage point off real consumption in a manageable scenario, and around 1–2 percent in a severe (i.e., outright CUSMA failure) scenario. Such a shock would show up quickly in discretionary and big-ticket purchases and then broaden if unemployment rises. Hence, a reasonable first scenario focusing on aggregate household consumption expenditure levels in Canada would be as described in Scenario 1 (or DFD1_HHLD_CONS).

Scenario 1: Overall household consumption expenditures in Canada are forecast to decline by 1 percent in short order (this scenario is referred to below as DFD1_HHLD_CON).

2.2 Agriculture, Agri-food and Seafood Exports to the U.S.

Current thinking reflected in much contemporary analysis and prognostication by Canada's private banks, industry associations, think-tanks, and various federal and provincial

---

[13] Scotiabank Economics (2026a). "Low Probability, High Cost: The Macroeconomics of Abandoning CUSMA." (https://www.scotiabank.com/ca/en/about/economics/economics-publications/post.other-publications.insights-views.cusma-scenario-note--march-5--2026-.html)

[14] As of May 30, 2026 Dr. Sylvain Charlebois of the Dalhousie Agri-Food Analytics Lab notes that Canada has recorded two consecutive quarters of negative real GDP growth leading many economists to say that Canada is now in a 'technical recession' – which is considerably below what the Bank of Canada (2025) projected (https://agrifoodanalyticslab.substack.com/p/canadas-economy-is-shrinking-the?utm_source=post-email-title&publication_id=1727895&post_id=199855826&utm_campaign=email-post-title&isFreemail=true&r=4boz42&triedRedirect=true&utm_medium=email). It is important to keep in mind that while Q4 2025 and Q1 2026 GDP growth figures were both negative, indicating a 'technical recession', these negative values were very small, and according to RBC Economics (May 29, 2026) "…the underlying details still…look firmer than what would be typically expected at, say, the beginning of a recession." For example, the report notes that consumer demand remained resilient in Q1 2026 with most of the Q1 26 GDP decline coming from a large drop in federal government weaponry purchases following several quarters of large increases.(https://www.rbc.com/en/economics/canadian-analysis/data-flashes/canadas-q1-gdp-posted-second-straight-although-small-decline/). RBC Economics (June 8, 2026) notes that "…the past two quarters of GDP decline both coincided with population declines – the first on record since the 1950s." As a result, this report notes that "[w]hile the total pie shrank, real per-person growth has continued to expand." They note that this is a better measure (i.e., per capital GDP growth) of how households and workers experience the economy. While headline GDP growth values may dictate a technical recession, per capita GDP growth values are more suggestive of Canada being in early stage recovery from what RBC calls a "soft patch" that started in early 2023. (https://www.rbc.com/en/economics/canadian-analysis/featured-analysis/insights/canadas-per-capita-economy-still-recovering/)

agencies holds that, at the mild end of the impact spectrum, with uncertainty effects only, agrifood exports to the U.S. can reasonably be expected to drop by between 0 and 5 percent (almost overnight). Under a more severe scenario, one where specific commodities are targeted, a 5-15 percent drop in affected categories (e.g., soy, processed meats and seafood) does not seem to be out of the question. In a case where both broad tariffs exist but CUSMA protections remain partially intact, a 10-20 percent decline appears reasonable. In a case of a severe tariff escalation such as the failure of CUSMA, a 20-35 percent drop in the export of processed food categories to the U.S. (especially prepared foods, beverages, bakery products, processed meat, packaged foods given their very thin margins, intense competition and easier substitution by U.S. suppliers) appears reasonable. Agri-Pulse (2026) has already noted that Canada to U.S. bilateral trade fell by 8-9 percent year-over-year from 2023 to 2024, and that in those product categories that were specifically targeted by retaliatory tariffs, declines of approximately 21 percent by value are expected.[15] Statistics Canada's Survey on Business conditions in 2025 revealed that 25.7 percent of exporters to the U.S. expected overall exports to the U.S. to decline under tariff conditions and for their financial conditions to worsen as a result. Indeed, Retail Insider (2025)[16] notes that a 25-35 percent tariff could make some agriculture and agrifood exports uneconomic, particularly those not subject to CUSMA exemptions (e.g., grains, oilseeds, processed foods and specialty value-added products) as these commodity groups are extremely sensitive to tariff changes given their very thin margins. Goods like canola oil and oil seeds, pork and beef products, greenhouse vegetables and specialty crops (80 percent of which flows to the U.S.) can reasonably be expected to experience declining export volumes on the order of anywhere from 5 to 20 percent. Not only will producers of these goods face lower demand, but the loss of a large market will also decrease their overall level of profitability. Even for products that do not cross the border, Canadian consumers are unlikely to be spared as processors who lose export markets attempt to recover their margins by increasing prices domestically - especially in those markets that are small. Tariffs will also wreak havoc upstream as input sourcing, contract logistics and production planning are all disrupted - all of this will make prices more volatile and cause sporadic availability of staple ingredients. Retaliatory tariffs will compound these effects via global supply chains (remember that Canadian food manufacturers depend on imported inputs including machinery, additives, packaging etc. which are also caught in this tariff crossfire). Concern exists over the effect of U.S. tariffs on

---

[15] Agripulse (2026). “Analysis: Checking in on US-Canada agricultural trade.” (https://www.agri-pulse.com/articles/23170-analysis-checking-in-on-us-canada-agricultural-trade?utm_source=chatgpt.com) – published May 15, 2026.

[16] Retail Insider (2025). “Canada’s Agri-Food Sector Hit by 35% U.S. Tariff After Ottawa Stalls.” (https://retail-insider.com/retail-insider/opinion/2025/08/canadas-agri-food-sector-hit-by-35-u-s-tariff-after-ottawa-stalls/?utm_source=chatgpt.com), August 1, 2025.

the relocation of major investments (from Canada to the U.S.) and the resulting reallocation of value-added production activities in various agrifood sectors. Overall, Canadian agri-food exports to the U.S. might decline on the order of 5-15 percent in a broad tariff environment with much larger losses in selected subsectors. Many sources (including RBC Economics, BoC, etc.) see agri-food as being more vulnerable to tariffs than farming itself and envision the possibility of Canada losing higher-value-added processing to the U.S. while retaining primary production. For seafood, no firm empirical predictions have been made (as far as this researcher is aware) but work by AgriPulse (2026) notes that overall bilateral trade in agriculture could decline by 8-9 percent with specifically targeted products declining by up to 21 percent. Given that seafood is similarly exposed to U.S. demand, and is likewise very price-elastic and perishable, a similar shock to seafood export demand would not be unreasonable. A reasonable second scenario focusing on the effect of U.S. tariffs on Canadian domestic exports of agriculture, agrifood and seafood products to the U.S. would be as described in Scenario 3 (or DFD3_AG_AF_SF).[17]

Scenario 3: Agriculture/Agri-food/Seafood exports to the U.S. (this scenario is referred to below as DFD3_AG_AF_SF)

	IO Industry No.	Industry Name	Export Shock[18]
1.	BS111A00	Crop Production	-8%
2.	BS1114A00	Greenhouse, Nursery and Floriculture	-15%
3.	BS112000	Animal Production and Aquaculture	-12%
4.	BS311400	Fruit and Vegetable Preserving/Specialty Food Manufacturing	-16%
5.	BS311200	Grain and Oilseed Milling	-12%
6.	BS311500	Dairy Product Manufacturing	-7%
7.	BS311600	Meat Production Manufacturing	-15%
8.	BS311700	Seafood Product Preparation and Packaging	-15%

2.4 Declining Steel and Aluminum Exports to the U.S.

The Financial Accountability Office of Ontario (FAO) (2025) notes that steel and aluminum exports to the U.S. could each decline by 25 percent by 2029. ING (2025) predicts a 22

[17] Note that in an earlier version of this report, there was a different 'second' scenario considered which has been subsequently dropped. In the interest of maintaining alignment with datasets and code, the labelling of the various scenarios has not been modified (e.g., this second scenario is actually labeled as "Scenario 3 or DFD3."

[18] Sub-sector-specific export shocks allocated regionally based on regional industry shares as of 2024.

percent drop in exports of Aluminum to the U.S. by late 2025. TD Economics (2026a) notes that Aluminum exports to the U.S. declined by 19 percent in Trump's first term (under less-punishing Section 232 tariffs). On this basis, a reasonable fourth scenario that focuses on the effect of U.S. tariffs on Canadian domestic exports of steel and aluminum to the U.S. would be as described in Scenario 4 (or DFD4_ST_AL).

Scenario 4: Steel and Aluminum Manufacturing and Manufactured Steel and Aluminum Products Exports to the U.S. Decline (this scenario is referred to as DFD4_ST_AL below.)

	IO Industry No.	Industry Name	Export Shock[19]
1.	BS331100	Iron and Steel Mills and Ferro-Alloy Manufacturing	-15%
2.	BS331300	Alumina and Aluminum Product Manufacturing	-10%

2.5 Softwood Lumber/Forestry Exports to the U.S.

RBC (2025a) notes that "[n]ew U.S. lumber tariffs … stack on top of pre-existing countervailing and anti-dumping (CTV/ATD) duties, leading to combined [tariff] rates exceeding 40% on Canadian softwood lumber exports…."[20] RBC (2025a) also notes that 50-60 percent of Canada's lumber production is exported to the U.S. and that geographic constraints associated with the resource, as well as economic structural constraints associated with the Canadian industry, leave few options for alternative markets for Canadian softwood lumber. Also relevant are forecasts of significant softening in homebuilding in the U.S. which stands to negatively impact U.S. demand for Canadian lumber. As of the start of 2025, RBC (2025a) notes that lumber production was already 20 percent lower than it was in 2021, and that it dropped an additional 4 percent in the first 3 quarters of 2024. So declining U.S. demand already translates into declining lumber production in Canada, and tariffs stand to further increase the price of Canadian lumber to U.S. firms. All of this suggests that a reasonable fifth scenario focused on the effect of U.S. tariffs on Canadian domestic exports of softwood lumber products would be as described in Scenario 5 (or DFD5_FOR).

Scenario 5: Softwood Lumber Exports to the U.S. Decline (this scenario is referred to as DFD5_FOR below)

[19] Sub-sector-specific export shocks allocated regionally based on regional industry shares as of 2024.
[20] RBC Economics (2025a). "Tracking the impact of U.S. tariffs on five targeted Canadian industries." (https://www.rbc.com/en/economics/canadian-analysis/featured-analysis/insights/tracking-the-impact-of-u-s-tariffs-on-five-targeted-canadian-industries/) .

	IO Industry No.	Industry Name	Export Shock[21]
1.	BS321100	Sawmills and Wood Preservation	-12%

2.6 Automobile and Automotive Parts Exports to the U.S.

The Ontario Financial Accountability Office (2025) notes that motor vehicle exports to the U.S. could decline by 20 percent and auto parts exports by 30%, by 2029. EDC (2025a and 2026a) predict that exports of motor vehicles and parts from Canada to the U.S. could decline by 4.3% in 2025 and by another 1.1% in 2026. All of this suggests that a reasonable sixth scenario focused on the effect of U.S. tariffs on Canadian domestic exports of automobiles and automotive parts would be as described in Scenario 6 (or DFD6_VEH_PTS).

Scenario 6: Automobile and Automotive Parts Exports to the U.S. Decline (this scenario is referred to as DFD6_VEH_PTS below).

	IO Industry No.	Industry Name	Export Shock[22]
1.	BS336100	Motor Vehicle Manufacturing	-20%
2.	BS336300	Motor Vehicle Parts Manufacturing	-15%

3.0 Measuring the Economic Impacts of Tariff Scenarios

The previous section describes five scenarios representing possible outcomes of the new U.S. tariffs on Canadian goods and services. A Multiregional Input-Output (MRIO) model of the Canadian economy, based on the most recent IO data for Canada and its regions (2022) was used to assess the impact of these scenarios across some 220 industrial sectors over 13 regions.[23] The MRIO model results were translated into impact metrics for each region including:

- Gross industry output impacts by industry
- Value-added impacts by industry
- Employment (in person-years) impacts by industry

Two versions of the MRIO model were implemented:

- An 'open' version which computes total economic impacts on industries that consist of direct and indirect output effects.

[21] Sub-sector-specific export shocks allocated regionally based on regional industry shares as of 2024.
[22] Sub-sector-specific export shocks allocated regionally based on regional industry shares as of 2024.
[23] Provincial Input-Output Tables, 2022. Catalogue 15-211-X.

    - Direct effects correspond to the 'first round' effects associated with a final demand shock (e.g., in order for the Iron and Steel Mills and Ferro-Alloy Manufacturing industry (BS331100) to produce one dollar of output, it must buy inputs from its suppliers including ores, energy, chemicals, shipping services, legal and financial services etc.. These sum to the direct effect of an additional dollar of output from the "Steel Mills and Ferro-Alloy Manufacturing" industry.)
    - Indirect effects pick up all rounds of spending that spill over from these direct input purchases by the industry in question (e.g., "Steel Mills and Ferro-Alloy Manufacturing" industry).
        - A big provider of inputs to this sector is the industry labelled "Iron ore mining" (BS212210). To increase output, the "Steel mills and ferro-alloy manufacturing" industry must purchase large amounts of the former's output (iron ore). This industry (i.e., Iron ore mining) must then also buy from its input suppliers to meet this demand (e.g., "Scientific research and development services" (BS541700), "Other professional, scientific and technical services" (BS541900), "Legal services" (BS541100) etc.). In turn, each of these input providers will have to draw on their input suppliers, and so on. The indirect effects are best thought of as the sum of spending rounds from 2 through n stimulated by the initial effect of increasing the output of "Steel mills and ferro-alloy manufacturing" by one dollar.
    - Use of an 'open' model involves the implicit assumption that all income earned by workers in satisfying all direct and indirect requirements stimulated by the initial shock does not re-enter the interindustry system via consumption (or other) demands.
- A 'closed' version – one that technically captures "Type II effects" between output, income, consumption and more production (i.e., where the total impact of a given shock is the sum of direct and indirect effects (as in the 'open model') as well as all 'induced' effects associated with the consumption expenditures stimulated by all earned income.

Capturing these Type II (i.e., induced) effects essentially involves splitting household consumption expenditures into two parts – an exogenous part that is affected by forces beyond a given interindustry system (e.g., by macroeconomic or political forces operating above the interindustry system such as a federal tax holiday or the sudden closure of a

foreign-owned auto manufacturing plant in response to changing tariff regimes) and an endogenous part that purchases inputs from and sells output to the other industries in that system as the system itself dictates (e.g., workers may use a portion of their annual income to purchase an automobile, to make securities investments (via a financial advisor), consume entertainment, pay local taxes, etc. and each of these purchases triggers additional (induced) demand for industrial outputs) in a manner established via fixed technical relationships between income, consumption and production.

3.1 National Impacts by Industry – 'Open Model' Results

Table 1 presents a summary of all impact metrics across all 13 provinces/territories, and for Canada as a whole, for the five tariff impact scenarios discussed above. For Canada as a whole, scenarios 6 (DFD6_VEH_PTS) and 1 (DFD1_AGG_CON) produce the largest economic impacts at the national scale. Scenario 6 (DFD6_VEH_PTS) posits export shocks of -20 percent and -15 percent in the two NAICS industries which underlay the heart of Canada's "automotive industry" (i.e., Motor Vehicle Manufacturing, and Motor Vehicle Parts Manufacturing respectively). This scenario translates into Canada-wide negative economic impacts in terms of gross output, value-added and employment of ($31.3B), ($9.6B) and -113,210 person-years respectively. These impacts are highly concentrated (more than 76%) in the Manufacturing industry (BS3A0) as shown in Figure 1.

Table 1: Impact Metric Summary (Open MRIO Model)

ID	Canada	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (13,852,190,120.01)	$ (25,901,065,088.24)	$ (13,646,800,316.17)	-149,069
3	DFD3_AG_AF_SF	$ (3,114,429,019.04)	$ (8,258,457,144.64)	$ (3,114,219,901.08)	-36,012
4	DFD4_ST_AL	$ (2,436,044,494.98)	$ (6,425,851,699.49)	$ (2,434,432,543.71)	-17,720
5	DFD5_FOR	$ (1,119,093,859.91)	$ (2,702,528,778.90)	$ (1,118,976,864.54)	-11,043
6	DFD6_VEH_PTS	$ (9,649,575,825.29)	$ (31,361,091,385.36)	$ (9,647,422,291.50)	-115,736
ID	NL	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (174,899,870.00)	$ (283,508,091.38)	$ (166,824,972.92)	-1,478
3	DFD3_AG_AF_SF	$ (50,747,746.31)	$ (130,501,315.83)	$ (58,299,295.10)	-600
4	DFD4_ST_AL	$ -	$ (9,942,175.03)	$ (7,385,317.59)	-19
5	DFD5_FOR	$ -	$ (14,429,578.35)	$ (11,086,488.62)	-22
6	DFD6_VEH_PTS	$ -	$ (22,291,420.07)	$ (15,191,785.49)	-63
ID	PEI	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (55,108,580.00)	$ (79,395,756.26)	$ (43,754,330.42)	-518
3	DFD3_AG_AF_SF	$ (74,617,040.10)	$ (143,492,864.94)	$ (54,884,811.15)	-780
4	DFD4_ST_AL	$ -	$ (554,498.03)	$ (254,523.19)	-3
5	DFD5_FOR	$ -	$ (691,245.88)	$ (312,916.81)	-4
6	DFD6_VEH_PTS	$ -	$ (6,157,333.05)	$ (2,334,740.33)	-31
ID	NS	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (347,152,350.00)	$ (523,470,679.98)	$ (285,646,563.30)	-3,420
3	DFD3_AG_AF_SF	$ (137,454,175.73)	$ (293,430,025.69)	$ (105,680,304.46)	-1,645
4	DFD4_ST_AL	$ -	$ (7,116,348.04)	$ (3,322,842.83)	-34
5	DFD5_FOR	$ (33,572,815.80)	$ (61,897,698.83)	$ (28,179,522.01)	-300
6	DFD6_VEH_PTS	$ -	$ (57,462,562.28)	$ (22,891,877.80)	-263
ID	NB	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (263,834,360.00)	$ (435,442,609.88)	$ (220,206,880.61)	-2,655
3	DFD3_AG_AF_SF	$ (88,119,517.86)	$ (231,147,089.24)	$ (75,650,770.65)	-1,087
4	DFD4_ST_AL	$ -	$ (10,972,216.85)	$ (3,759,011.44)	-41
5	DFD5_FOR	$ (67,145,631.59)	$ (136,072,064.13)	$ (53,892,285.63)	-607
6	DFD6_VEH_PTS	$ -	$ (44,856,145.17)	$ (17,682,191.37)	-203
ID	QC	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (2,765,970,680.00)	$ (5,314,440,974.48)	$ (2,682,232,104.00)	-33,594
3	DFD3_AG_AF_SF	$ (433,619,145.59)	$ (1,165,635,660.72)	$ (411,162,863.57)	-6,428
4	DFD4_ST_AL	$ (1,295,642,864.24)	$ (2,917,588,192.66)	$ (1,261,736,890.09)	-7,122
5	DFD5_FOR	$ (257,391,587.78)	$ (588,591,178.55)	$ (233,345,901.03)	-2,615
6	DFD6_VEH_PTS	$ (964,957,582.53)	$ (3,966,968,357.40)	$ (1,539,792,529.22)	-22,994
ID	ON	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (5,438,832,450.01)	$ (10,571,294,459.49)	$ (5,394,753,827.51)	-60,507
3	DFD3_AG_AF_SF	$ (683,695,227.59)	$ (1,843,068,520.91)	$ (720,195,035.24)	-9,609
4	DFD4_ST_AL	$ (861,021,939.71)	$ (2,459,379,404.60)	$ (7,498.91)	-7,499
5	DFD5_FOR	$ (89,527,508.79)	$ (311,421,706.35)	$ (127,059,580.29)	-1,290
6	DFD6_VEH_PTS	$ (8,684,618,242.76)	$ (25,918,006,836.86)	$ (7,301,061,805.40)	-87,691
ID	MB	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (461,972,410.00)	$ (832,920,507.61)	$ (455,241,666.42)	-4,627
3	DFD3_AG_AF_SF	$ (497,692,970.03)	$ (1,160,647,639.99)	$ (408,260,652.70)	-4,421
4	DFD4_ST_AL	$ -	$ (33,064,914.10)	$ (16,879,172.26)	-128
5	DFD5_FOR	$ (11,190,938.60)	$ (32,063,778.30)	$ (13,612,712.76)	-141
6	DFD6_VEH_PTS	$ -	$ (173,536,435.73)	$ (88,734,019.40)	-694
ID	SK	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (413,386,270.00)	$ (803,351,381.14)	$ (449,290,383.62)	-4,122
3	DFD3_AG_AF_SF	$ (434,078,278.19)	$ (1,150,578,455.21)	$ (482,925,294.60)	-2,610
4	DFD4_ST_AL	$ -	$ (45,260,628.43)	$ (21,856,380.60)	-120
5	DFD5_FOR	$ (11,190,938.60)	$ (35,955,682.19)	$ (17,057,584.28)	-109
6	DFD6_VEH_PTS	$ -	$ (153,434,862.33)	$ (93,824,375.34)	-396
ID	AB	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (1,716,382,490.00)	$ (3,535,370,836.19)	$ (1,915,420,951.13)	-17,028
3	DFD3_AG_AF_SF	$ (544,089,309.91)	$ (1,700,122,014.88)	$ (610,021,321.88)	-5,890
4	DFD4_ST_AL	$ (66,232,456.90)	$ (353,550,781.29)	$ (175,009,029.34)	-980
5	DFD5_FOR	$ (134,291,263.19)	$ (409,638,271.56)	$ (189,388,960.04)	-1,228
6	DFD6_VEH_PTS	$ -	$ (665,025,209.06)	$ (363,278,992.97)	-1,901
ID	BC	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (2,168,945,990.00)	$ (3,455,388,667.11)	$ (1,996,196,295.99)	-20,853
3	DFD3_AG_AF_SF	$ (170,315,607.72)	$ (436,353,426.57)	$ (185,550,968.48)	-2,933
4	DFD4_ST_AL	$ (213,147,234.13)	$ (584,517,327.37)	$ (175,505,944.71)	-1,763
5	DFD5_FOR	$ (514,783,175.56)	$ (1,111,070,838.21)	$ (444,669,658.60)	-4,725
6	DFD6_VEH_PTS	$ -	$ (325,559,310.89)	$ (187,359,940.36)	-1,437
ID	YT	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (17,147,840.00)	$ (23,653,764.52)	$ (14,043,442.10)	-105
3	DFD3_AG_AF_SF	$ -	$ (463,997.20)	$ (244,636.91)	-2
4	DFD4_ST_AL	$ -	$ (1,086,746.81)	$ (584,212.21)	-1
5	DFD5_FOR	$ -	$ (210,574.69)	$ (109,345.26)	-1
6	DFD6_VEH_PTS	$ -	$ (7,899,405.59)	$ (4,242,381.93)	-20
ID	NT	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (19,049,220.00)	$ (25,234,076.21)	$ (13,641,844.05)	-95
3	DFD3_AG_AF_SF	$ -	$ (750,241.85)	$ (400,794.05)	-2
4	DFD4_ST_AL	$ -	$ (431,595.92)	$ (219,381.92)	-1
5	DFD5_FOR	$ -	$ (293,530.63)	$ (155,036.48)	-1
6	DFD6_VEH_PTS	$ -	$ (2,286,556.06)	$ (1,134,842.16)	-6
ID	NU	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (9,507,610.00)	$ (17,593,283.98)	$ (9,547,054.08)	-67
3	DFD3_AG_AF_SF	$ -	$ (2,265,891.63)	$ (943,152.31)	-6
4	DFD4_ST_AL	$ -	$ (2,386,870.35)	$ (1,340,960.57)	-5
5	DFD5_FOR	$ -	$ (192,631.22)	$ (106,872.73)	-1
6	DFD6_VEH_PTS	$ -	$ (17,606,950.87)	$ (9,892,809.74)	-38

Figure 1

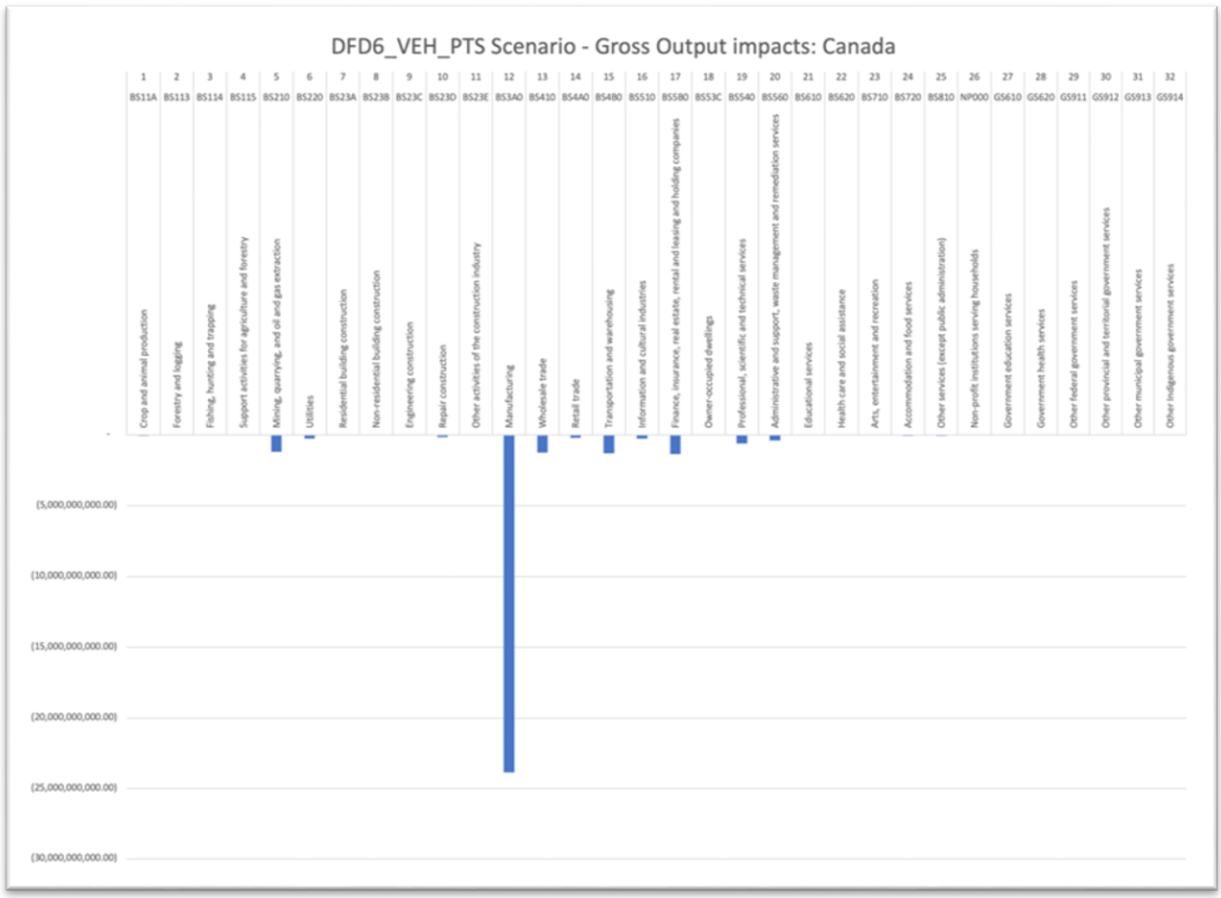


Figure 2 presents the impacts associated with DFD6_VEH_PTS at a more detailed level of industrial detail (where BS3A0 is broken into 70 sub-sectors).[24] It shows that the impact to "Manufacturing" in Figure 1 to be concentrated largely in the "Automobile and light duty truck manufacturing" (BS336110) and the "Motor vehicle parts manufacturing" (BS336300) industries. Strong backward linkages to several other "manufacturing" industries including "Iron and steel mills and ferro-alloy manufacturing" (BS331200), "Plastic product manufacturing" (BS326100), "Non-ferrous metal (except aluminum) production and processing" (BS331400), and "Computer and other electronic product manufacturing" (BS334C00) – industries which house firms involved in the heart of Canada's manufacturing core - are clear to see. While Figures 1 and 2 pertain to Canada as whole,

[24] The "Manufacturing" sector shown in Figures 1 through 4 (or BS3A0 – equivalent to NAICS 31 through 33). BS3A0 is the aggregate Input-Output Industry Code (IOIC) used by Statistics Canada in its GDP accounting system). This aggregate manufacturing sector appears as one industry in the "Summary-Level" IO tables but in the "Detail Level" tables, this is broken into 70 more finely grained sub-sectors.

the focus on the automotive and parts manufacturing (and linked) industries suggests a clear geography to these impacts (see below for more on the regional distribution of impacts).

Figure 2:

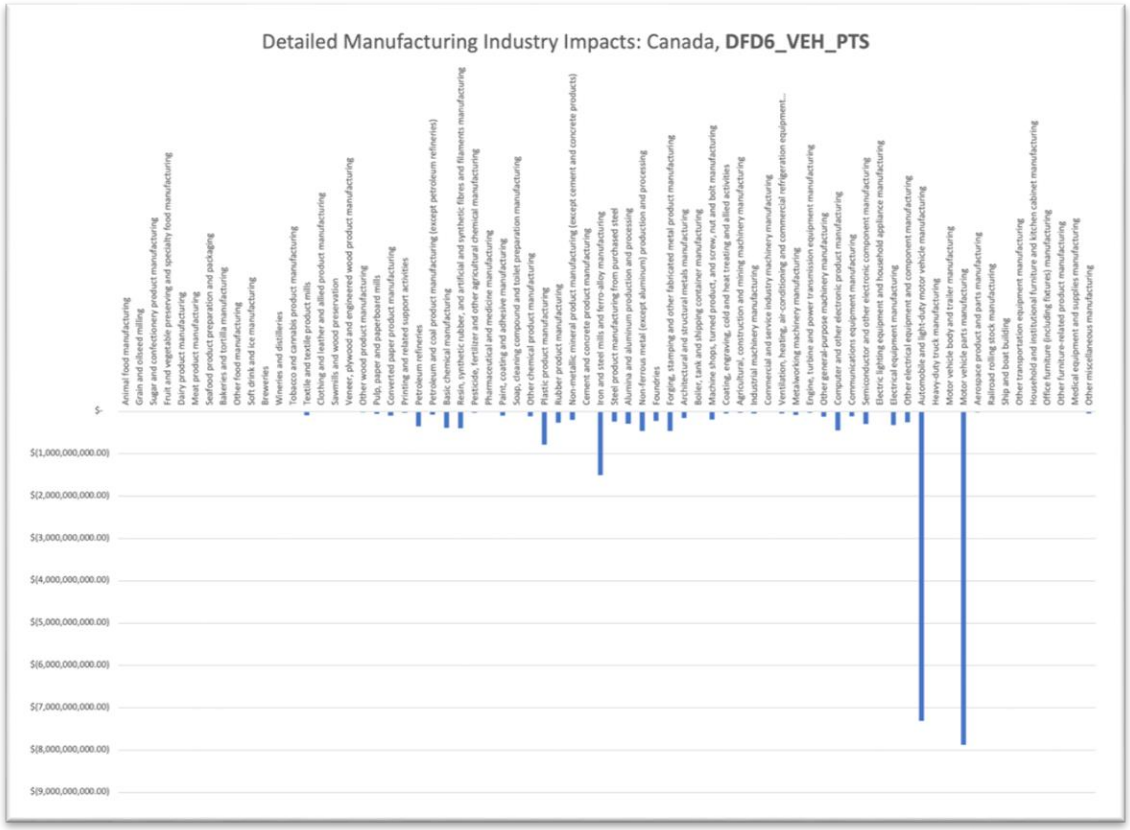
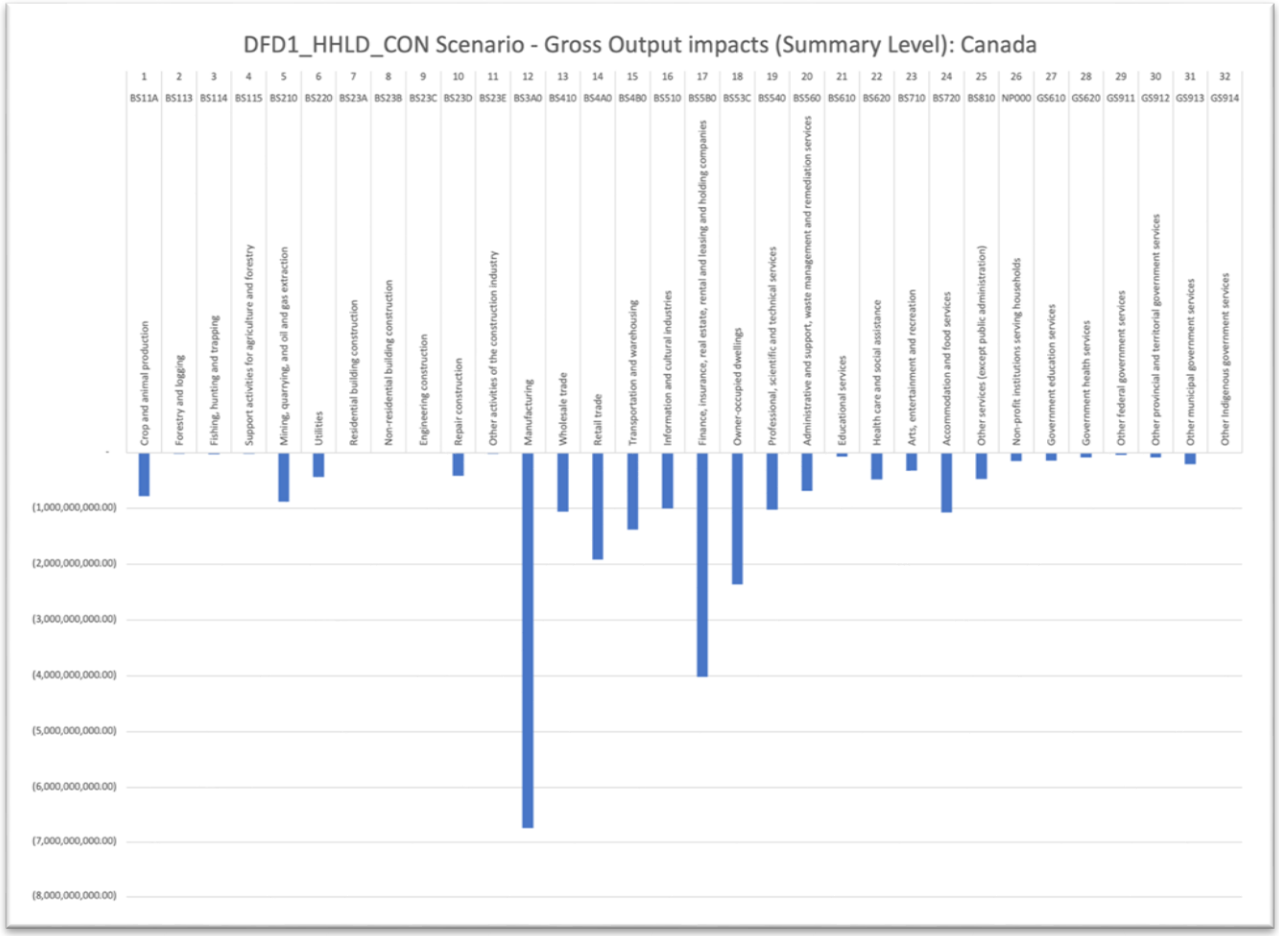


Table 1 shows that Scenario 1 (DFD1_HHLD_CON – which posits a 1% decline in aggregate household consumption expenditures) ranks second in terms of gross industry output, value-added and employment impacts at the national scale. This is not surprising given that household consumption expenditures generate nearly 60 percent of Canadian GDP. Figure 3 shows that such a shock would generate far-reaching (in terms of the breadth of industries impacted and, as is discussed below, across regions) repercussions for the Canadian economy.

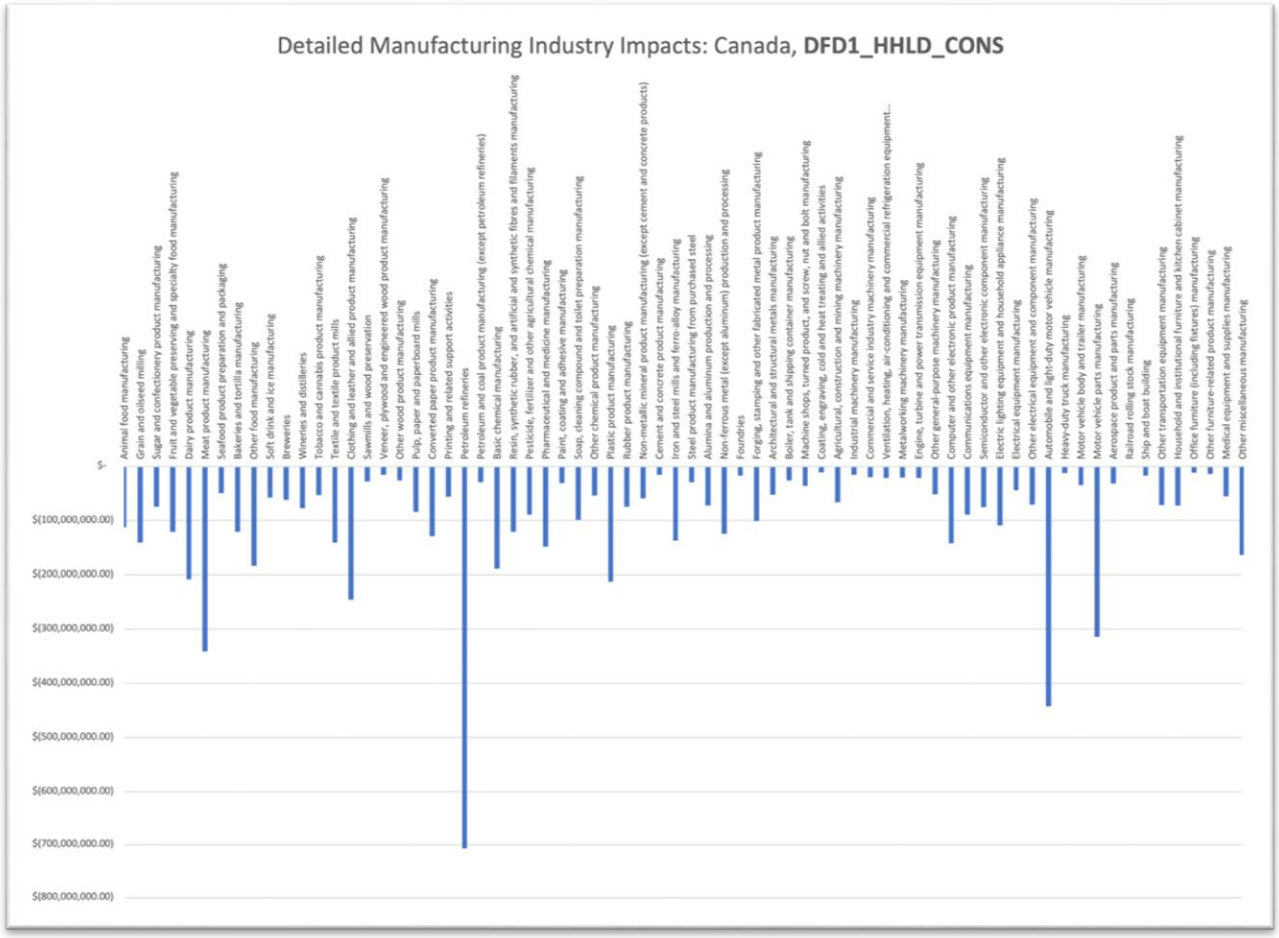
Figure 3:



Indeed, Figure 3 shows that the gross output impacts (and indeed the value-added and employment effects) associated with a 1% decline in household consumption expenditures (DFD1_HHLD_CON) accrue to a far wider range of industries than was the case in Figures 1 and 2 (which pertain to DFD6_VEH_PTS) – ranging from Manufacturing (BS3A0), to “Finance, insurance, real estate and rental and leasing and holding companies” (BS5B0), to “Retail trade” (BS4A0), “Accommodation and food services” (BS720), and all the way back to “Crop and animal production” (BS11A).

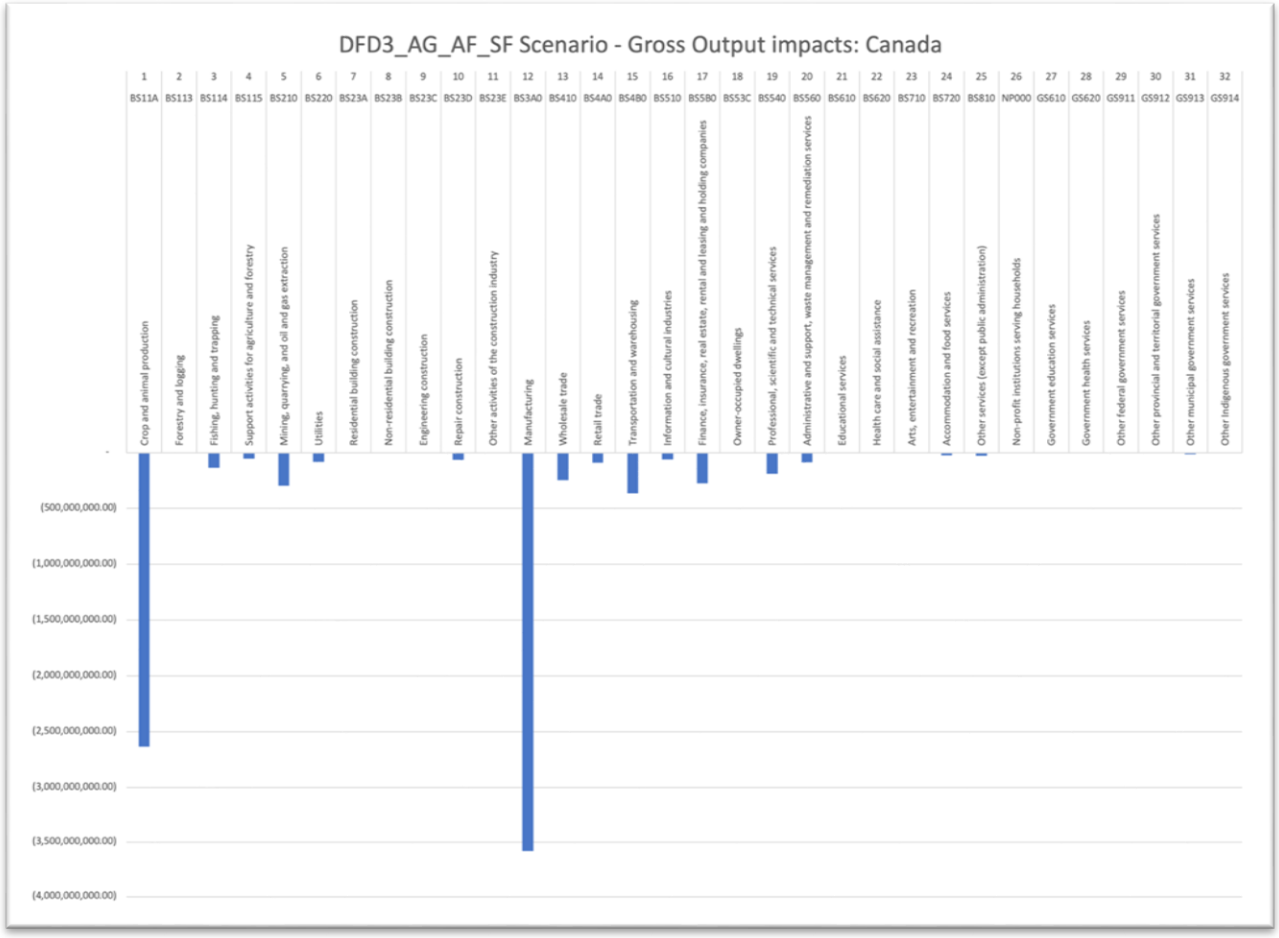
Figure 4 shows that the “Manufacturing” sector impacts associated with DFD1_HHLD_CON shown in Figure 3 generates a more broad-spectrum economic impact nationally than does DFD6_VEH_PTS.

Figure 4:



The third-ranked scenario in terms of gross output, value-added and employment impact at the national scale is DFD3_AG_AF_SF which posits significant export shocks to agriculture, agri-food and seafood industries in Canada – industries which, to varying degrees, have been deemed highly vulnerable to tariffs (not just from the U.S.). Many of these goods, especially the live products including cattle, hogs, and crustaceans, are overwhelmingly dependent on the U.S. market and have little potential for market diversification. As such, U.S. tariffs have the potential to greatly impact these industries. Indeed, it has been suggested that with these industries that typically operate with razor-thin margins, even low tariff levels could be disastrous, especially so for those activities which involve processing (i.e., agri-food). Figure 5 presents the national economic impacts associated with DFD3_AG_AF_SF.

Figure 5:



Evident in Figure 5 is a very strong connection between “Crop and animal production” (BS11A – which, at this level of sectoral aggregation, includes aquaculture and aspects of seafood production) and “Manufacturing” (BS3A0).

Figure 6:

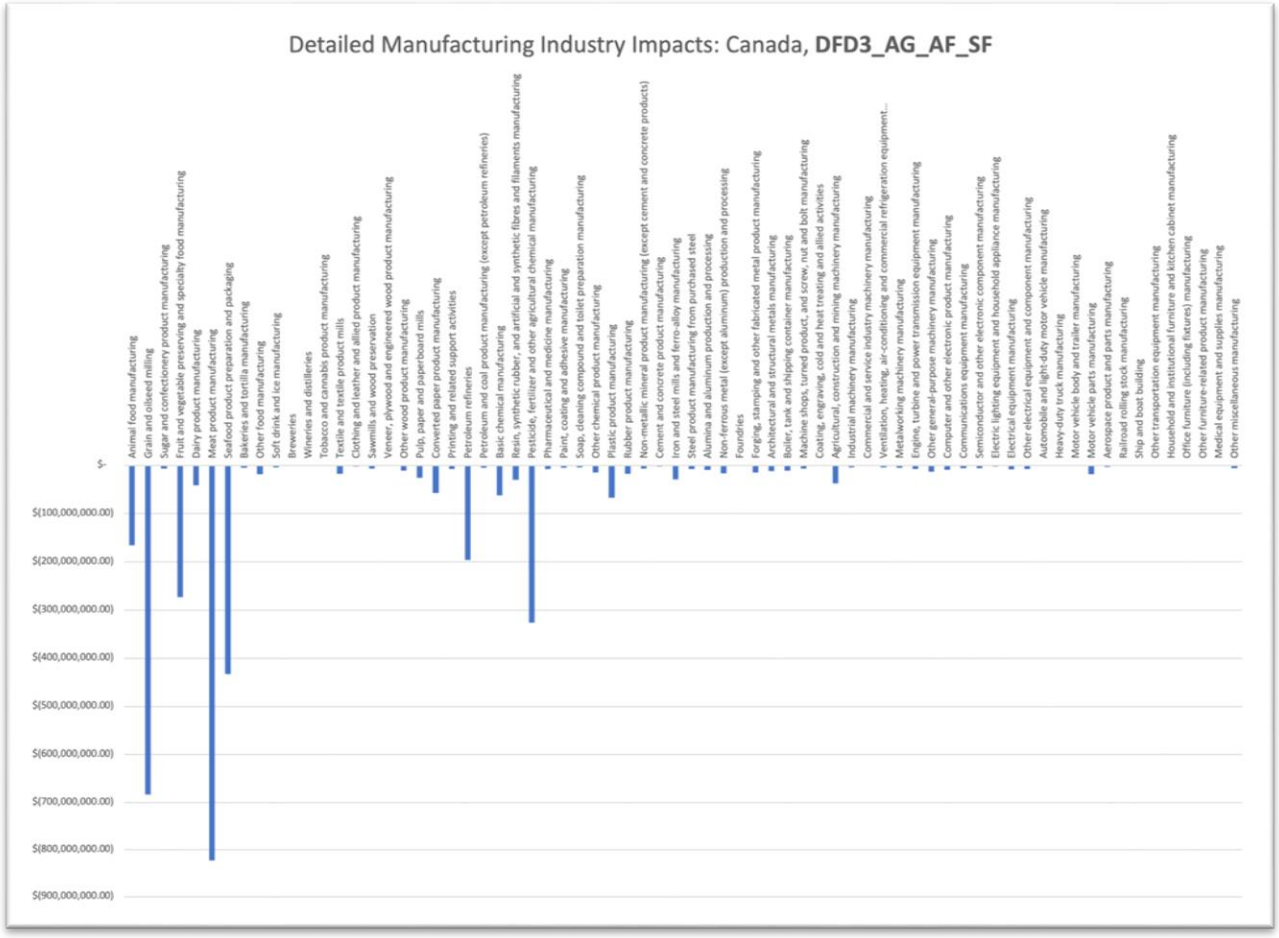


Figure 6 reveals both backward linkages (i.e., from agriculture, agri-food and seafood to their input suppliers including primary, secondary and tertiary inputs) and forward linkages (i.e., to those industries which use their outputs as input including “Seafood product preparation and packaging” (BS311700)).

Figure 7:

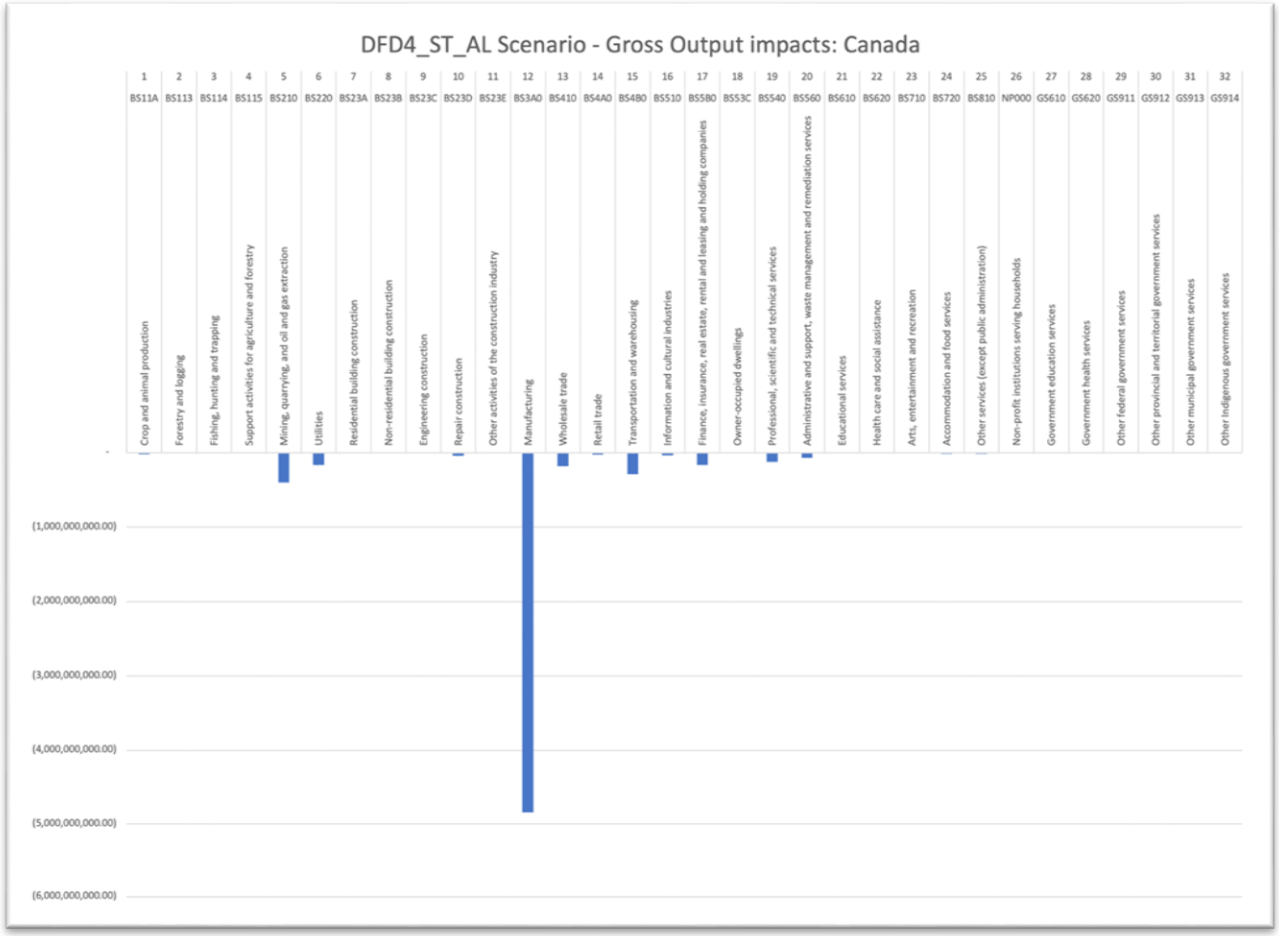


Figure 7 presents this same information for DFD4_ST_AL (steel and aluminum manufacturing), which ranks fourth in terms of impact magnitudes Canada-wide. Again, manufacturing dominates the impact profile followed at a considerable distance by “Finance, Insurance, Real Estate and Leasing and Holding Companies” (BS5B0), “Transportation and Warehousing” (BS4B0), “Wholesale Trade” (BS410) and “Mining, Quarrying and Oil and Gas Extraction “(BS210) industries.

Figure 8

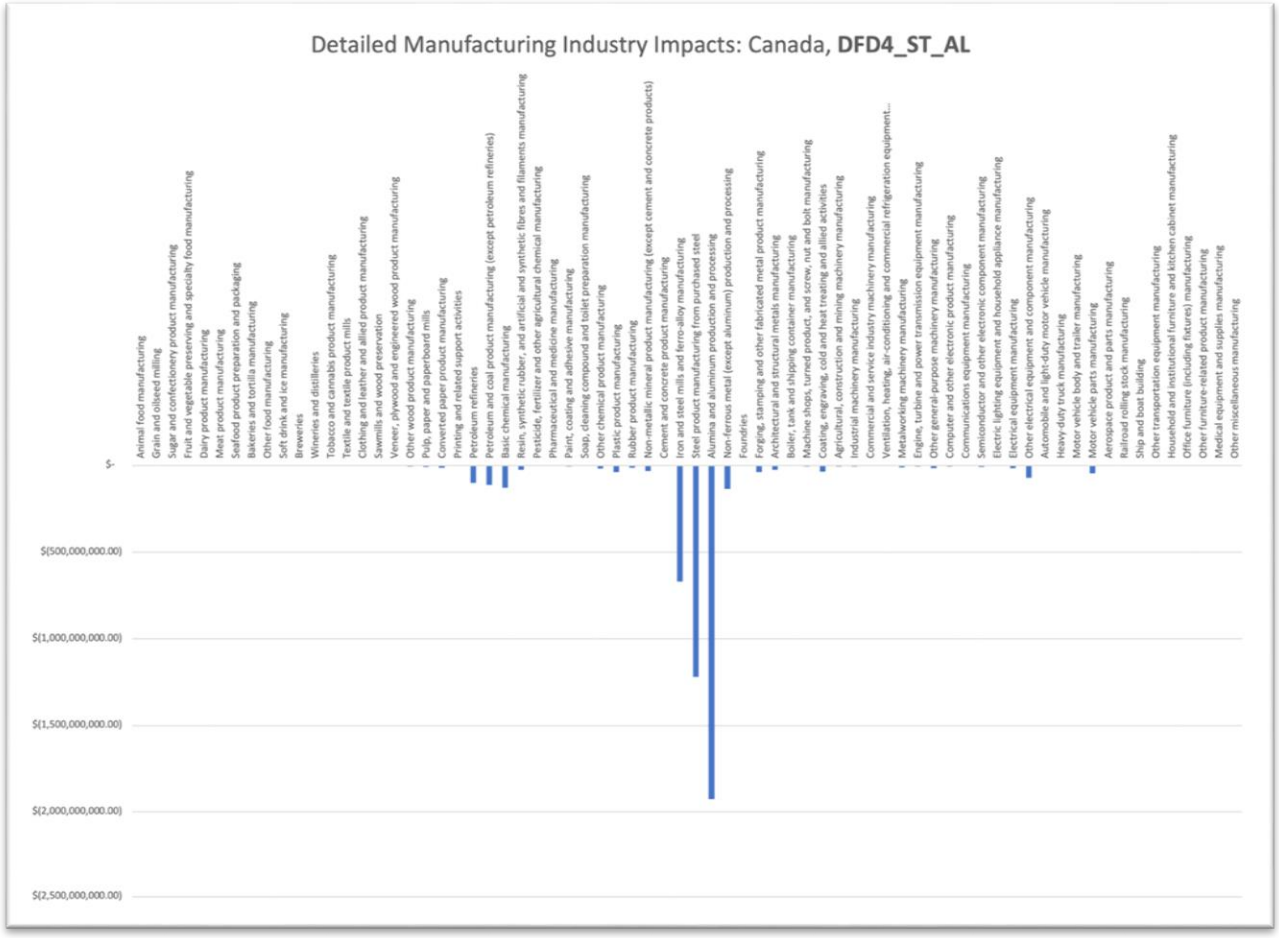


Figure 8 reveals that the economic impacts to manufacturing in DFD4_ST_AL consist largely of impacts to “Alumina and aluminum production and processing” (BS331300), “Steel product manufacturing from purchased steel” (BS331200), and “Iron and steel mills and ferro-alloy manufacturing” (BS331100) industries.

Figure 9

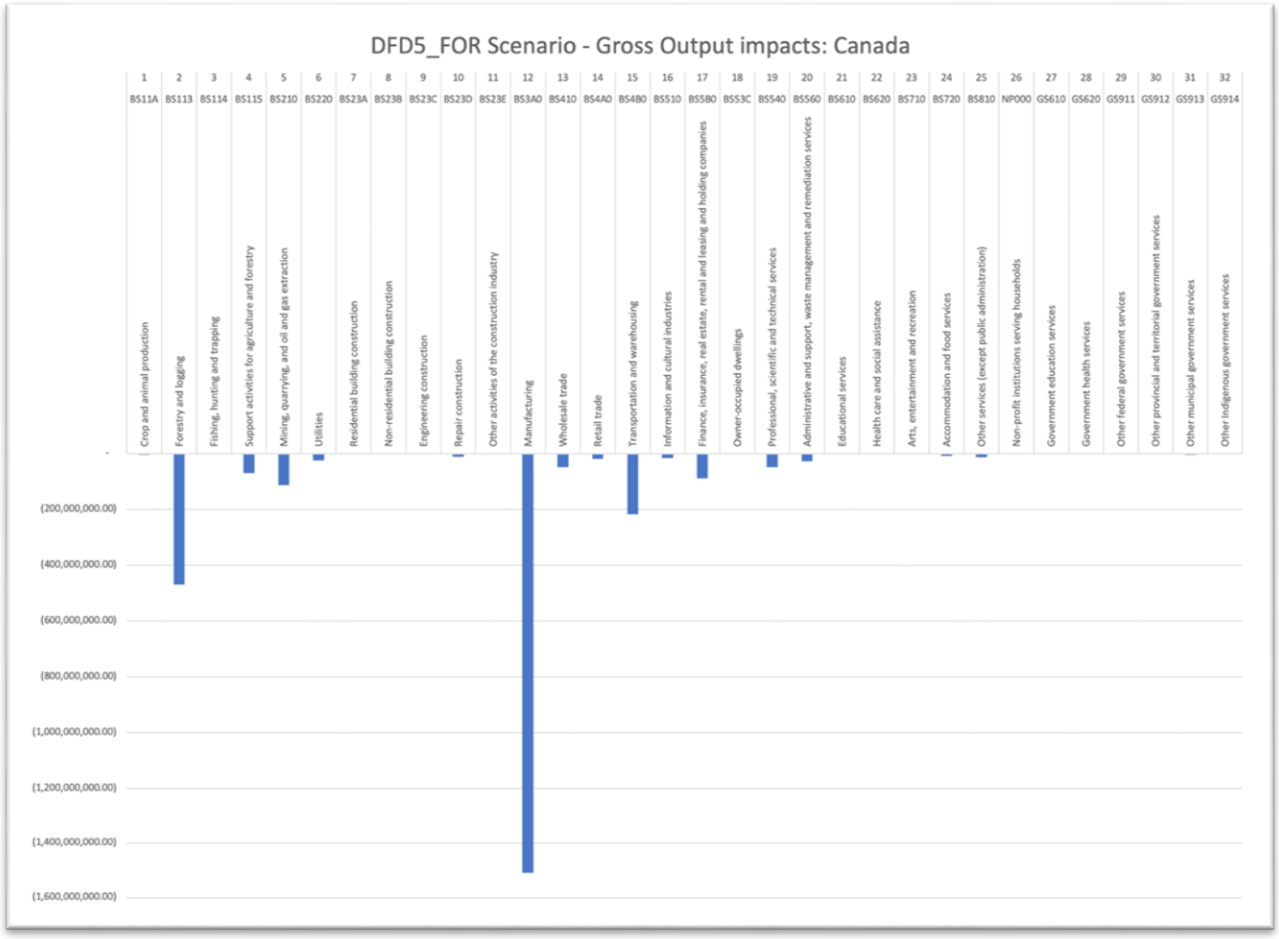


Figure 9 presents the impact profile associated with the final and least impactful scenario considered – DFD5_FOR which consists of a -12 percent shock in the export of softwood lumber to the U.S. Figure 9 shows that this reduction in softwood lumber exports to the U.S. also results in a disproportionate impact to “Manufacturing” (BS3A0), followed at a considerable distance by “Forestry and logging (BS114) and “Transportation and warehousing” (BS4B0). Many other service sectors, including “Finance, insurance, real estate, rental and leasing and holding companies” (BS5B0) are also impacted.

Figure 10:

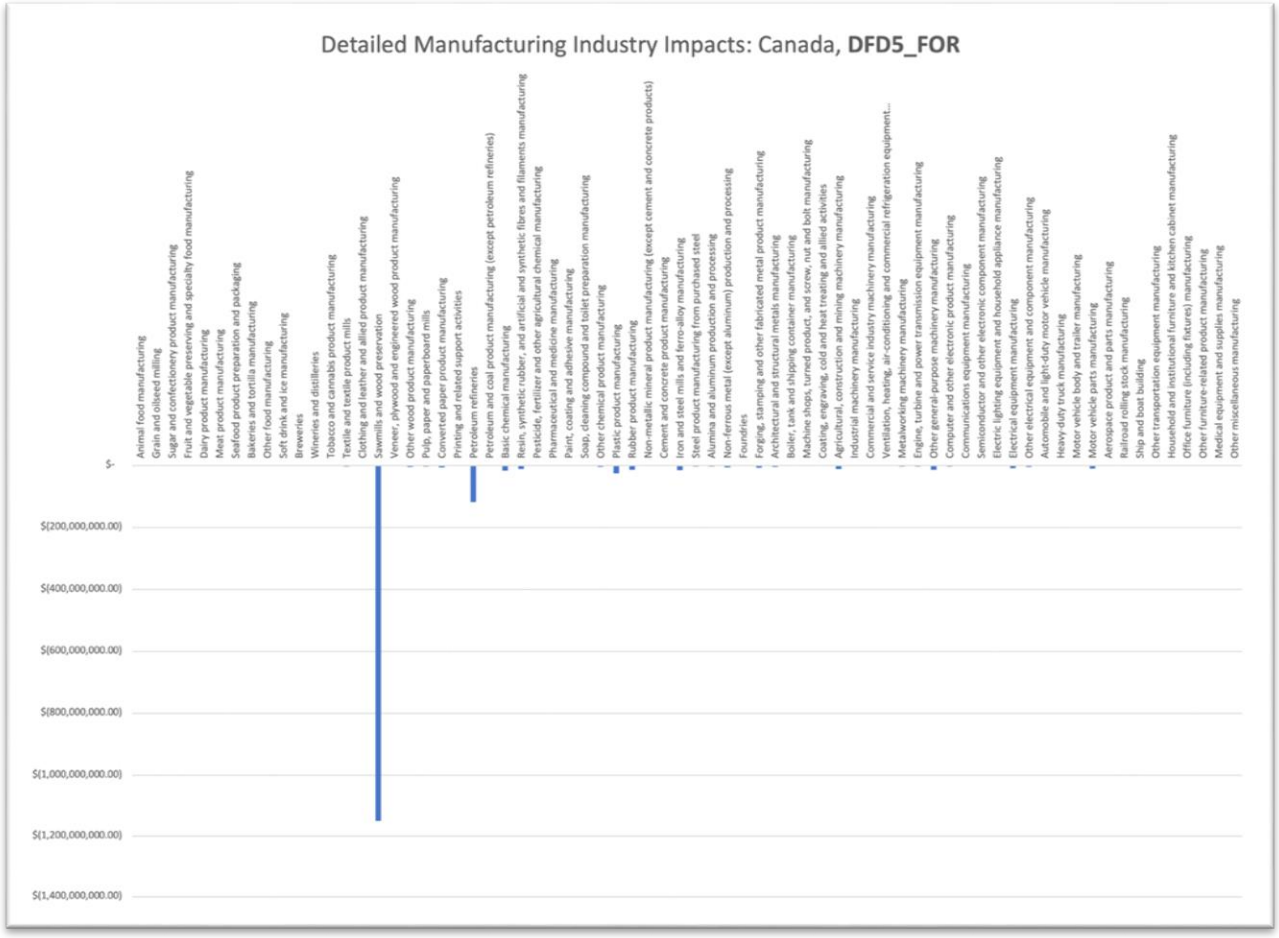


Figure 10 examines the manufacturing sector impact associated with DFD5_FOR in more detail and shows that one manufacturing sub-sector, “Sawmills and wood preservation” (BS321100), accounts for much of this impact.

Across all five scenarios, notwithstanding tremendous differences in scale and activities, the pervasiveness of a significant manufacturing sector impact is noteworthy. This finding underscores the reality that all the industries targeted in the various scenarios will have widespread economic impacts via the strong connection to domestic and highly diverse manufacturing industries which suggests that the resulting induced effects stand to be very significant.

3.2 National Impacts by Industry – 'Closed Model' Results

Table 2 shows that, like in the open MRIO model results discussed above, the first and fifth scenarios (DFD1_HHLD_CON and DFD5_VEH_PTS) still generate the largest Canada-wide economic impacts of the five scenarios under consideration. As discussed above, interpretation of economic impacts generated by an open model involves the implicit assumption that household income earned (lost) in response to interindustry effects of exogenous final demand shocks (in this case, the responses to the 5 tariff scenarios) does not re-enter the interindustry system by way of endogenous consumption effects. The closed model represents an attempt to endogenize that portion of household consumption expenditure behaviour that is tied to production-related household income. So, while a given scenario may engender output responses across 220 industries in 13 regions in the open model case, this output response is inclusive of direct and indirect effects alone. In addition to these direct and indirect effects, the closed MRIO model captures what are called 'induced effects' which represent the additional output (or value-added, employment, etc.) effects that are attributable to the additional (or lost) household consumption expenditures associated with the labour income earned (lost) because of the direct and indirect effects of a given scenario. Not surprisingly then, the impacts generated by a closed model are always greater than are those generated by an open model for a given shock. It is also the case that these induced effects can alter the regional distribution of total impacts as regions differ in terms of their roles in the satisfaction of household consumption demand.[25]

Comparing Tables 1 and 2 reveals that the closed MRIO model does indeed generate larger impacts in terms of gross industry output, value-added and person-years of employment, for the same tariff shocks, relative to the open MRIO model. Comparing Figures 2 and 11 (i.e., detailed manufacturing sector impacts associated with DFD6_VEH_PTS generated by open and closed MRIO models respectively) does allow one to appreciate the contribution of induced effects associated with the shock to motor vehicle and motor vehicle parts manufacturing industries. For one, Figure 11 shows that each industrial impact is larger than its counterpart in Figure 2 which excludes the induced effects. While both open and closed model outputs show that the two key automotive industries garner the lion's share

[25] For example, labour income earned by workers in Alberta's oil patch may lead allow some of these workers to purchase new automobiles (e.g., new Ford F-150s or Honda Accords, or Toyota Camrys, etc.) As such, the additional wages paid to workers in Alberta's oil patch could increase final demand for the output of the "Automobile and light-duty motor vehicle manufacturing" industry (BS336100) of Ontario. This is an example of an induced output effect felt by BS336100 in Ontario resulting from changes in activity levels experienced in the oil and gas industry of Alberta. The open model misses this entirely – hence the closed model will produce a qualitatively different regional distribution of impacts relative to the open model for a given final demand shock.

of the total impact of scenario DFD6_VEH_PTS, Figure 11 does show that several more industries (relative to Figure 2) are impacted under the scenario when the closed MRIO model is employed. Indeed, gross output, value-added and employment impacts extend much further into the primary and tertiary sectors when induced effects are included, as compared to open model case where they are not.

Figure 11:

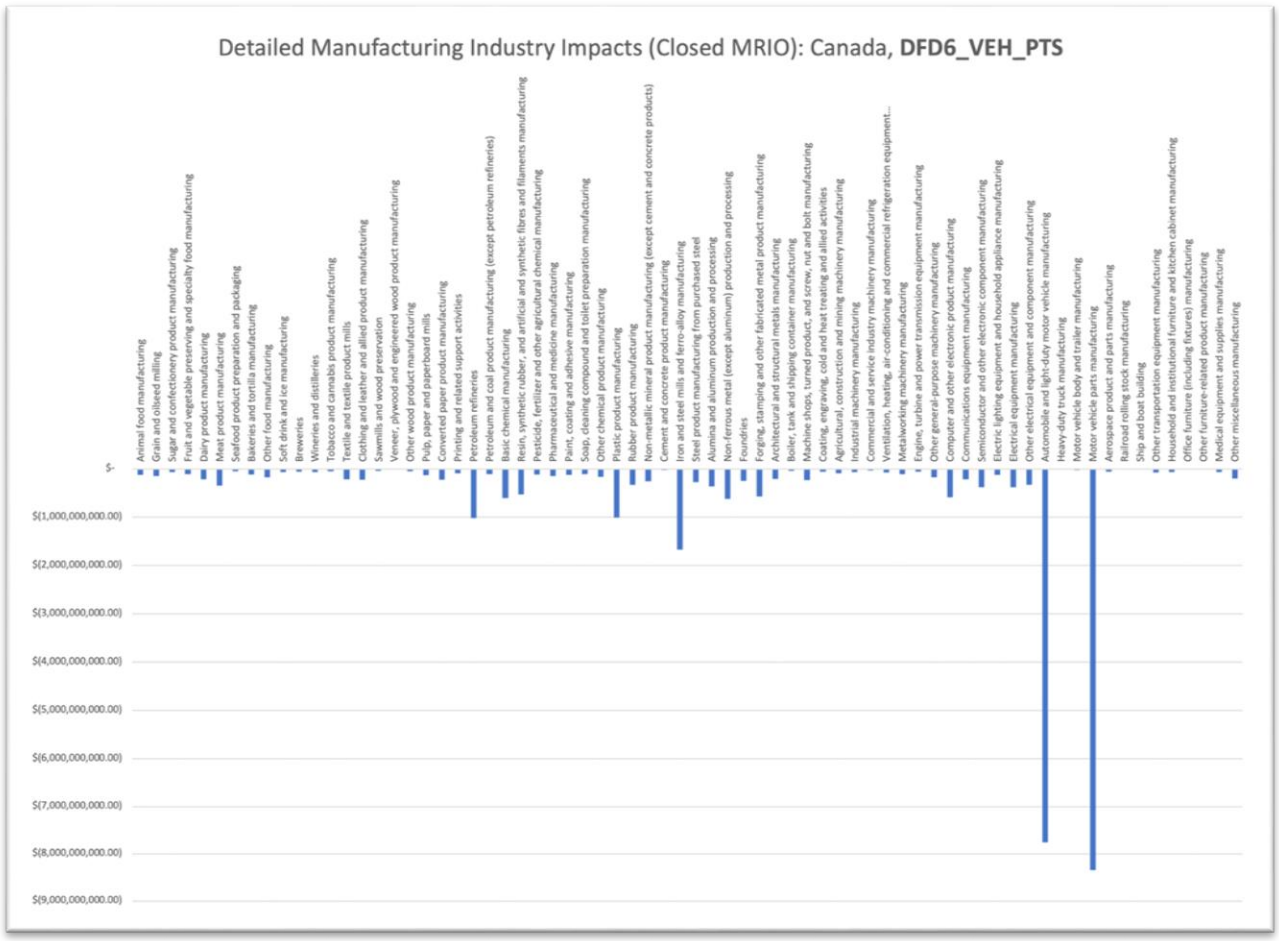


Table 2: Impact Metric Summary (Closed MRIO Model)

ID	Canada	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (13,852,190,120.01)	$ (61,417,876,962.62)	$ (32,262,424,536.76)	-349,762
3	DFD3_AG_AF_SF	$ (3,114,429,019.04)	$ (14,720,345,465.00)	$ (6,513,632,946.82)	-72,362
4	DFD4_ST_AL	$ (2,436,044,494.98)	$ (10,905,057,795.06)	$ (4,760,569,336.22)	-43,320
5	DFD5_FOR	$ (1,119,093,859.91)	$ (4,840,140,665.23)	$ (2,254,353,053.56)	-23,174
6	DFD6_VEH_PTS	$ (9,649,575,825.29)	$ (56,245,678,883.65)	$ (22,547,817,110.66)	-256,382
ID	NL	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (174,899,870.00)	$ (546,932,816.36)	$ (339,432,737.50)	-2,754
3	DFD3_AG_AF_SF	$ (50,747,746.31)	$ (205,519,226.54)	$ (107,855,917.53)	-974
4	DFD4_ST_AL	$ -	$ (27,356,439.50)	$ (18,981,059.37)	-84
5	DFD5_FOR	$ -	$ (25,081,516.76)	$ (18,391,786.13)	-62
6	DFD6_VEH_PTS	$ -	$ (107,364,754.42)	$ (69,591,667.10)	-397
ID	PEI	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (55,108,580.00)	$ (159,918,930.76)	$ (89,498,973.75)	-1,059
3	DFD3_AG_AF_SF	$ (74,617,040.10)	$ (198,244,973.86)	$ (88,585,556.97)	-1,159
4	DFD4_ST_AL	$ -	$ (4,827,242.82)	$ (2,446,373.19)	-31
5	DFD5_FOR	$ -	$ (3,573,972.14)	$ (1,791,255.80)	-23
6	DFD6_VEH_PTS	$ -	$ (32,023,246.67)	$ (15,369,368.25)	-199
ID	NS	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (347,152,350.00)	$ (1,044,537,216.52)	$ (582,800,053.68)	-6,975
3	DFD3_AG_AF_SF	$ (137,454,175.73)	$ (452,817,517.64)	$ (198,574,816.23)	-2,745
4	DFD4_ST_AL	$ -	$ (31,258,275.10)	$ (15,712,345.22)	-192
5	DFD5_FOR	$ (33,572,815.80)	$ (92,680,302.23)	$ (45,613,589.30)	-509
6	DFD6_VEH_PTS	$ -	$ (204,708,052.96)	$ (99,563,652.78)	-1,241
ID	NB	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (263,834,360.00)	$ (894,756,246.99)	$ (445,331,356.91)	-5,341
3	DFD3_AG_AF_SF	$ (88,119,517.86)	$ (363,388,411.52)	$ (141,311,874.00)	-1,862
4	DFD4_ST_AL	$ -	$ (41,724,479.05)	$ (17,151,796.87)	-208
5	DFD5_FOR	$ (67,145,631.59)	$ (186,462,357.76)	$ (80,890,523.02)	-921
6	DFD6_VEH_PTS	$ -	$ (197,510,896.71)	$ (84,994,295.01)	-1,041
ID	QC	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (2,765,970,680.00)	$ (12,448,204,526.89)	$ (6,317,123,189.66)	-77,251
3	DFD3_AG_AF_SF	$ (433,619,145.59)	$ (2,343,331,746.51)	$ (1,005,062,221.54)	-13,538
4	DFD4_ST_AL	$ (1,295,642,864.24)	$ (4,363,423,153.10)	$ (2,016,345,331.19)	-16,231
5	DFD5_FOR	$ (257,391,587.78)	$ (1,023,874,308.17)	$ (455,972,738.95)	-5,286
6	DFD6_VEH_PTS	$ (964,957,582.53)	$ (8,524,034,886.82)	$ (3,848,246,009.93)	-50,766
ID	ON	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (5,438,832,450.01)	$ (27,150,609,062.94)	$ (13,673,155,677.77)	-153,135
3	DFD3_AG_AF_SF	$ (683,695,227.59)	$ (4,463,854,851.01)	$ (2,004,092,678.18)	-24,002
4	DFD4_ST_AL	$ (861,021,939.71)	$ (4,459,016,673.36)	$ (18,638.18)	-18,638
5	DFD5_FOR	$ (89,527,508.79)	$ (986,508,949.95)	$ (447,904,444.11)	-4,907
6	DFD6_VEH_PTS	$ (8,684,618,242.76)	$ (42,328,895,957.21)	$ (15,756,961,467.04)	-181,879
ID	MB	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (461,972,410.00)	$ (1,889,702,731.10)	$ (1,039,206,875.80)	-10,448
3	DFD3_AG_AF_SF	$ (497,692,970.03)	$ (1,620,618,137.40)	$ (675,821,891.57)	-7,142
4	DFD4_ST_AL	$ -	$ (111,017,740.99)	$ (57,675,631.24)	-521
5	DFD5_FOR	$ (11,190,938.60)	$ (75,368,586.27)	$ (36,407,355.98)	-372
6	DFD6_VEH_PTS	$ -	$ (645,936,771.94)	$ (339,046,202.84)	-3,028
ID	SK	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (413,386,270.00)	$ (1,668,473,382.02)	$ (944,552,079.08)	-8,130
3	DFD3_AG_AF_SF	$ (434,078,278.19)	$ (1,520,104,406.54)	$ (701,121,022.35)	-4,576
4	DFD4_ST_AL	$ -	$ (110,985,587.44)	$ (58,563,335.74)	-382
5	DFD5_FOR	$ (11,190,938.60)	$ (71,998,440.16)	$ (36,557,326.33)	-263
6	DFD6_VEH_PTS	$ -	$ (527,725,773.11)	$ (308,494,369.44)	-1,759
ID	AB	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (1,716,382,490.00)	$ (7,864,086,907.97)	$ (4,289,187,389.38)	-37,285
3	DFD3_AG_AF_SF	$ (544,089,309.91)	$ (2,613,381,783.81)	$ (1,114,284,385.77)	-10,320
4	DFD4_ST_AL	$ (66,232,456.90)	$ (760,303,791.16)	$ (396,122,901.21)	-2,731
5	DFD5_FOR	$ (134,291,263.19)	$ (687,224,481.86)	$ (338,876,199.22)	-2,521
6	DFD6_VEH_PTS	$ -	$ (2,416,653,780.55)	$ (1,313,682,318.69)	-8,971
ID	BC	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (2,168,945,990.00)	$ (7,620,573,371.98)	$ (4,465,945,879.57)	-46,856
3	DFD3_AG_AF_SF	$ (170,315,607.72)	$ (929,127,253.03)	$ (471,637,237.01)	-6,011
4	DFD4_ST_AL	$ (213,147,234.13)	$ (986,540,347.21)	$ (413,127,372.18)	-4,274
5	DFD5_FOR	$ (514,783,175.56)	$ (1,684,700,203.30)	$ (790,444,429.30)	-8,301
6	DFD6_VEH_PTS	$ -	$ (1,200,125,079.52)	$ (677,329,487.42)	-6,917
ID	YT	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (17,147,840.00)	$ (47,359,599.00)	$ (29,375,769.82)	-214
3	DFD3_AG_AF_SF	$ -	$ (2,235,012.30)	$ (1,313,035.94)	-10
4	DFD4_ST_AL	$ -	$ (2,516,496.85)	$ (1,463,401.59)	-4
5	DFD5_FOR	$ -	$ (858,435.03)	$ (502,764.65)	-4
6	DFD6_VEH_PTS	$ -	$ (17,736,196.05)	$ (10,278,743.08)	-61
ID	NT	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (19,049,220.00)	$ (44,490,771.08)	$ (25,121,680.15)	-170
3	DFD3_AG_AF_SF	$ -	$ (2,575,218.86)	$ (1,398,593.13)	-8
4	DFD4_ST_AL	$ -	$ (1,566,619.17)	$ (847,163.15)	-5
5	DFD5_FOR	$ -	$ (899,183.41)	$ (487,101.06)	-3
6	DFD6_VEH_PTS	$ -	$ (9,430,710.88)	$ (5,080,999.54)	-30
ID	NU	Shock	Delta-X	Delta-VA	Delta-Jobs
1	DFD1_HHLD_CON	$ (9,507,610.00)	$ (38,231,399.00)	$ (21,692,873.71)	-145
3	DFD3_AG_AF_SF	$ -	$ (5,146,925.99)	$ (2,573,716.61)	-16
4	DFD4_ST_AL	$ -	$ (4,520,949.31)	$ (2,585,364.98)	-13
5	DFD5_FOR	$ -	$ (909,928.18)	$ (513,539.69)	-3
6	DFD6_VEH_PTS	$ -	$ (33,532,776.80)	$ (19,178,529.54)	-94

3.3 Provincial Distribution of Gross Output/Value-Added Impacts

Figure 12 shows the composition of the total gross output impacts associated with the most impactful scenario considered, DFD6_VEH_PTS, across the 13 regions of Canada as estimated by the open MRIO model. The fact that all scenarios, especially DFD6_VEH_PTS reveal significant direct and indirect (and in the case of the closed MRIO, induced) effects to Manufacturing which is heavily concentrated in the Province of Ontario, shows up clearly in Figure 12. Clearly, the majority of this impact will accrue to the Province of Ontario.

Figure 12:

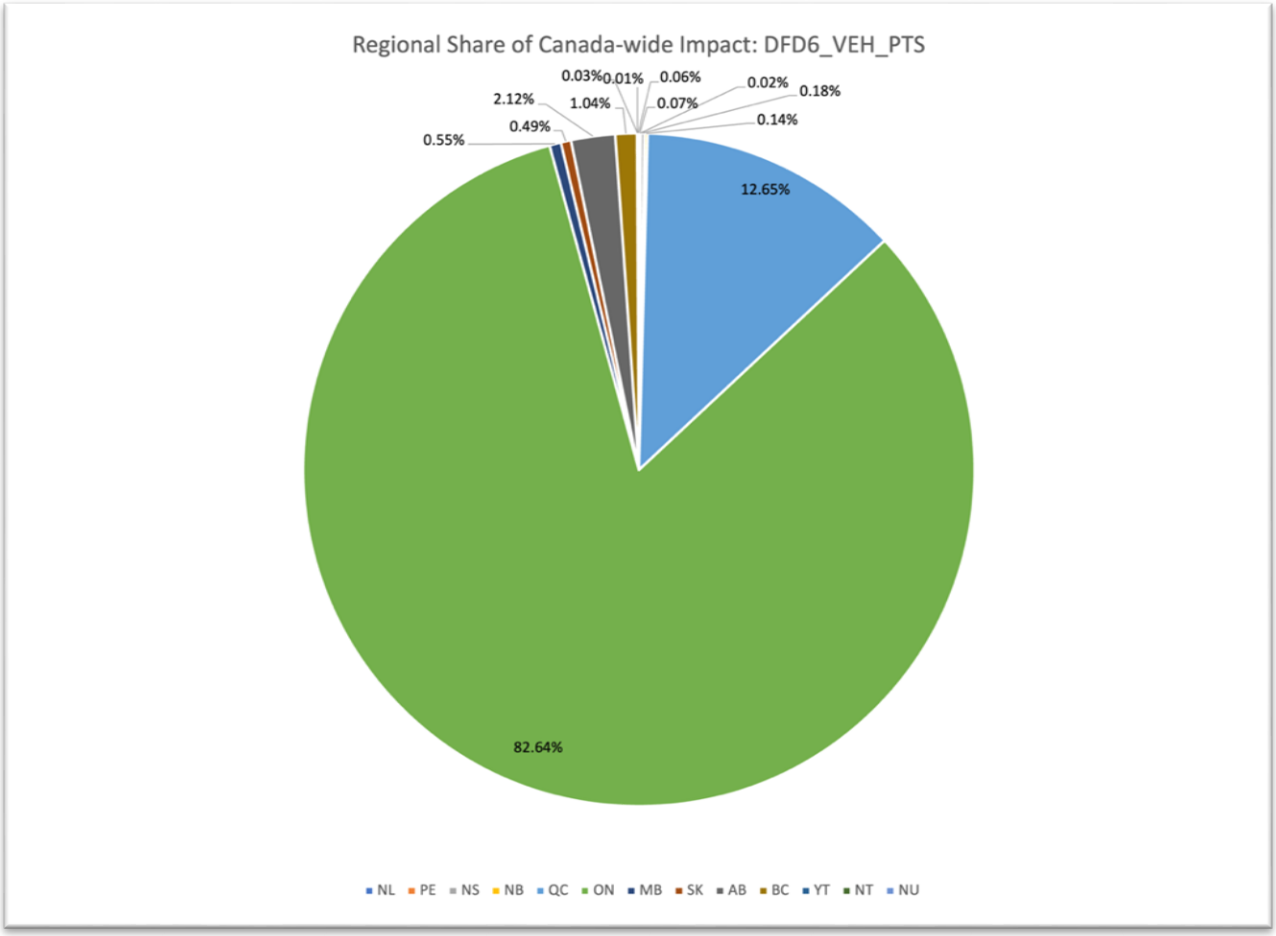


Figure 13 presents this same information but for the results of the closed MRIO model. As noted earlier, the inclusion of induced effects in the closed model results act to adjust both the sectoral and the spatial distribution of the economic impacts associated with DFD6_VEH_PTS. Specifically, Figure 13 shows that the economic impacts remain concentrated in the Province of Ontario when induced impacts are included, but to a slightly lesser extent, with other regions, notably the Province of Quebec, increasing their share of the total impact of DFD6_VEH_PTS.

Figure 13:

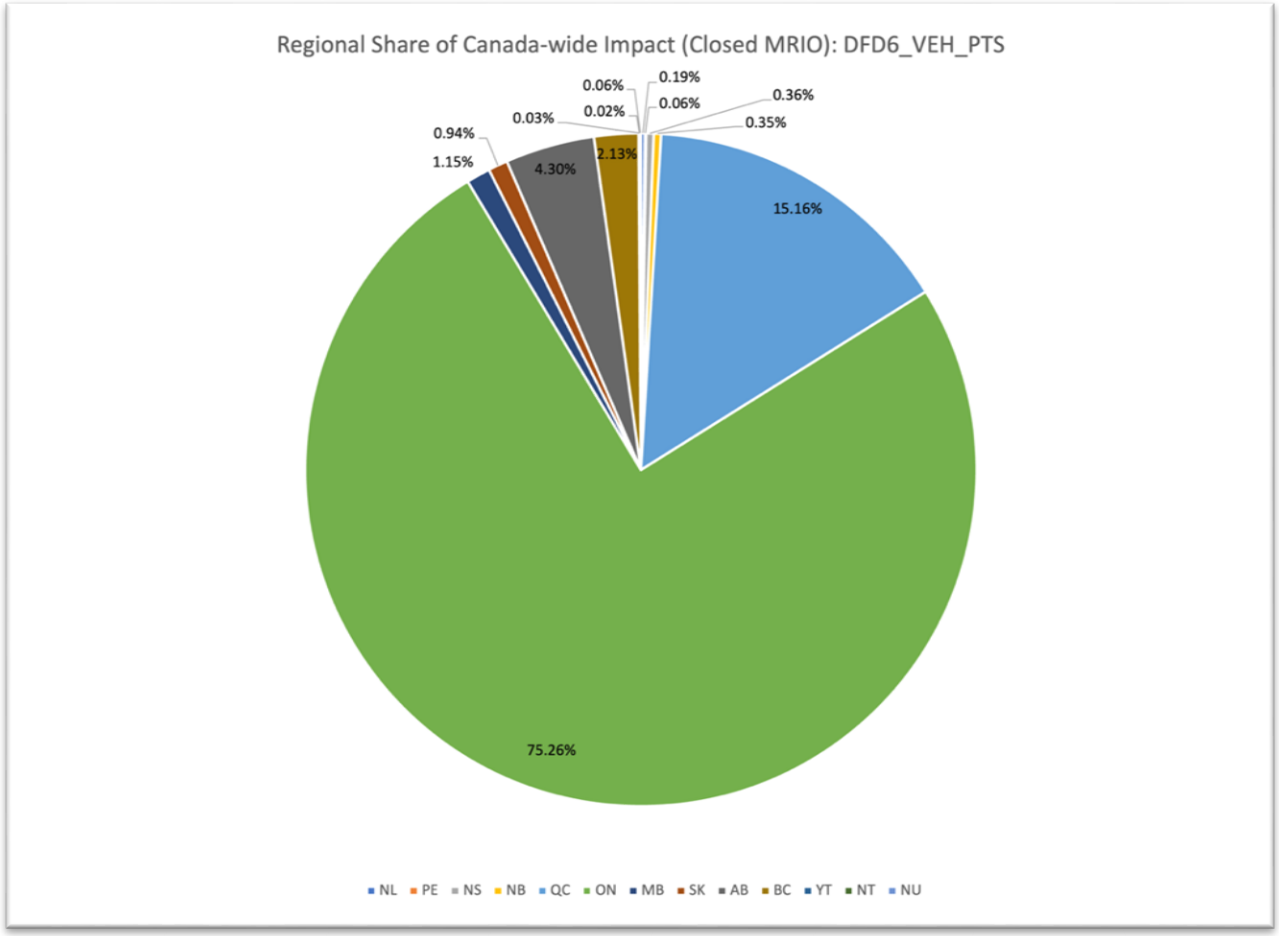
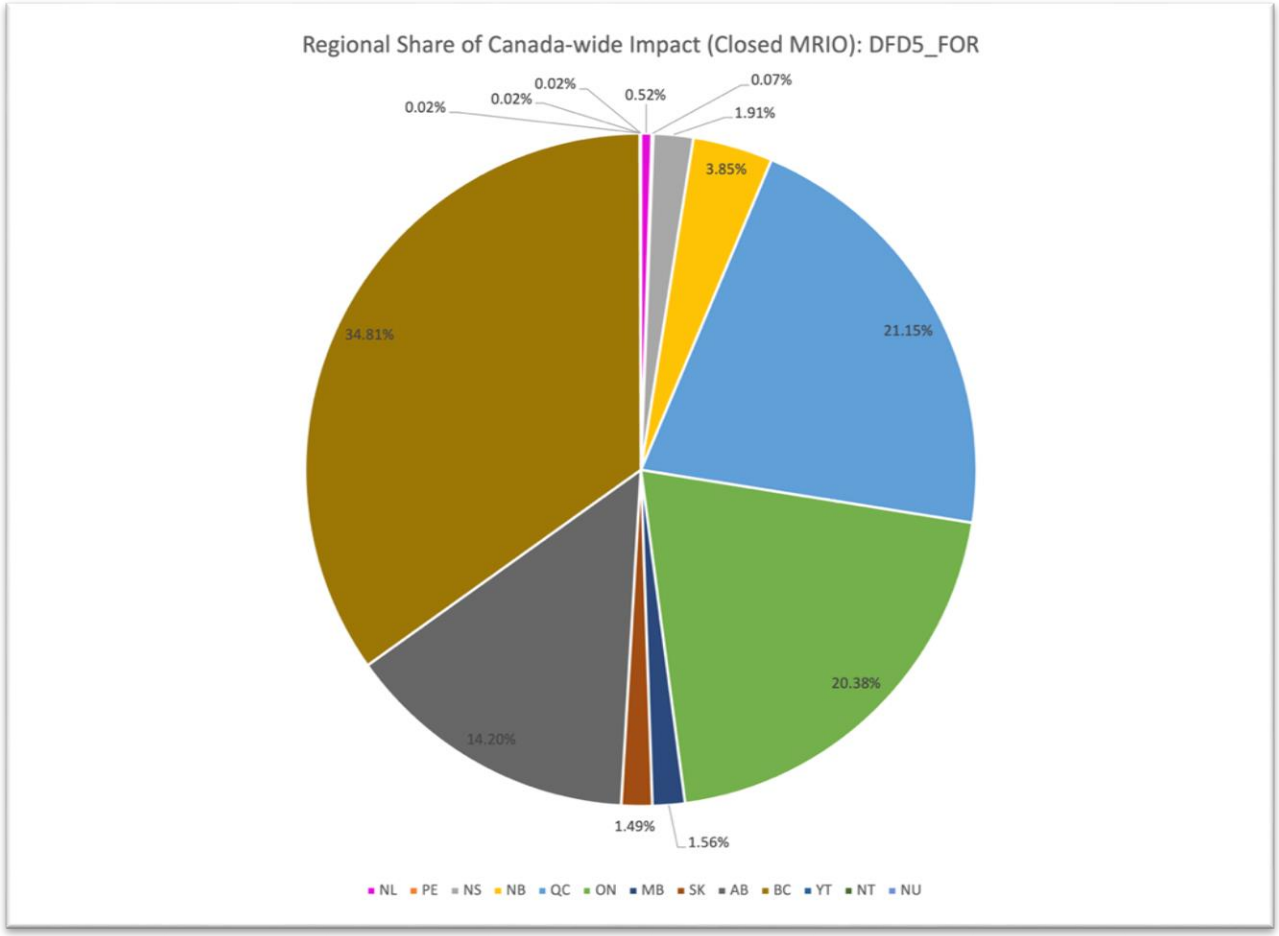

Regional Share of Canada-wide Impact (Closed MRIO): DFD6_VEH_PTS
0.06%
0.19%
0.36%
0.06%
0.03%
0.02%
0.35%
0.94%
2.13%
1.15%
4.30%
15.16%
75.26%
NL PE NS NB QC ON MB SK AB BC YT NT NU

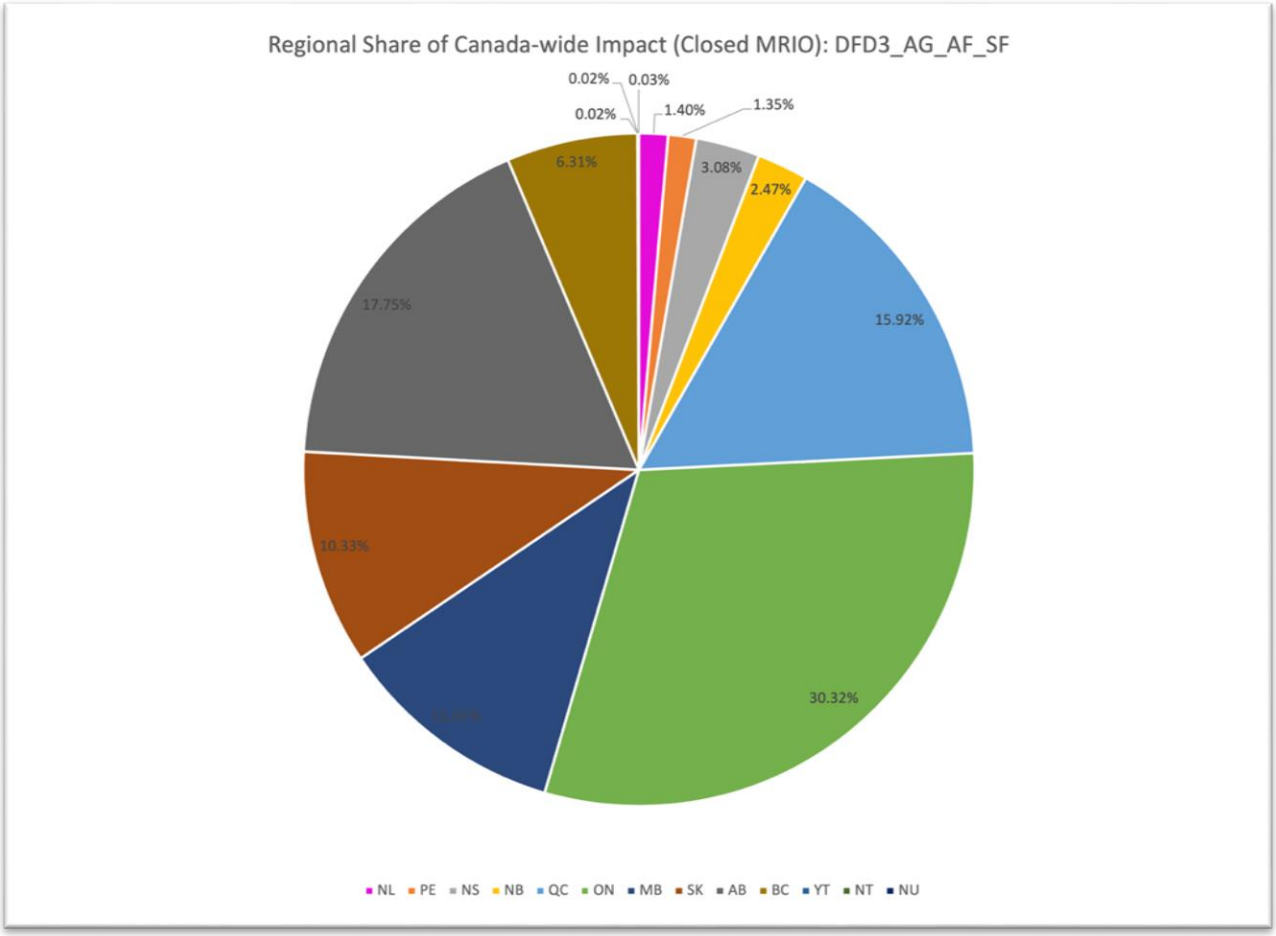
Figure 14:



In contrast, Figure 14 shows that DFD5_FOR (which posits the reduction in softwood lumber exports to the U.S.), while directly impacting the regions of Canada which are representative of this industry (i.e., the Provinces of British Columbia, Quebec and Alberta) generates nearly equally sizable economic impacts in those regions which represent Canada's manufacturing heartland (i.e., the Provinces of Ontario and Quebec).

Figure 15 shows that DFD3_AG_AF_SF (which posits reductions in several primary industries and their aligned processing industries) the impacts, especially those including induced effects, are widely and slightly more evenly distributed across the regions of Canada.

Figure 15:

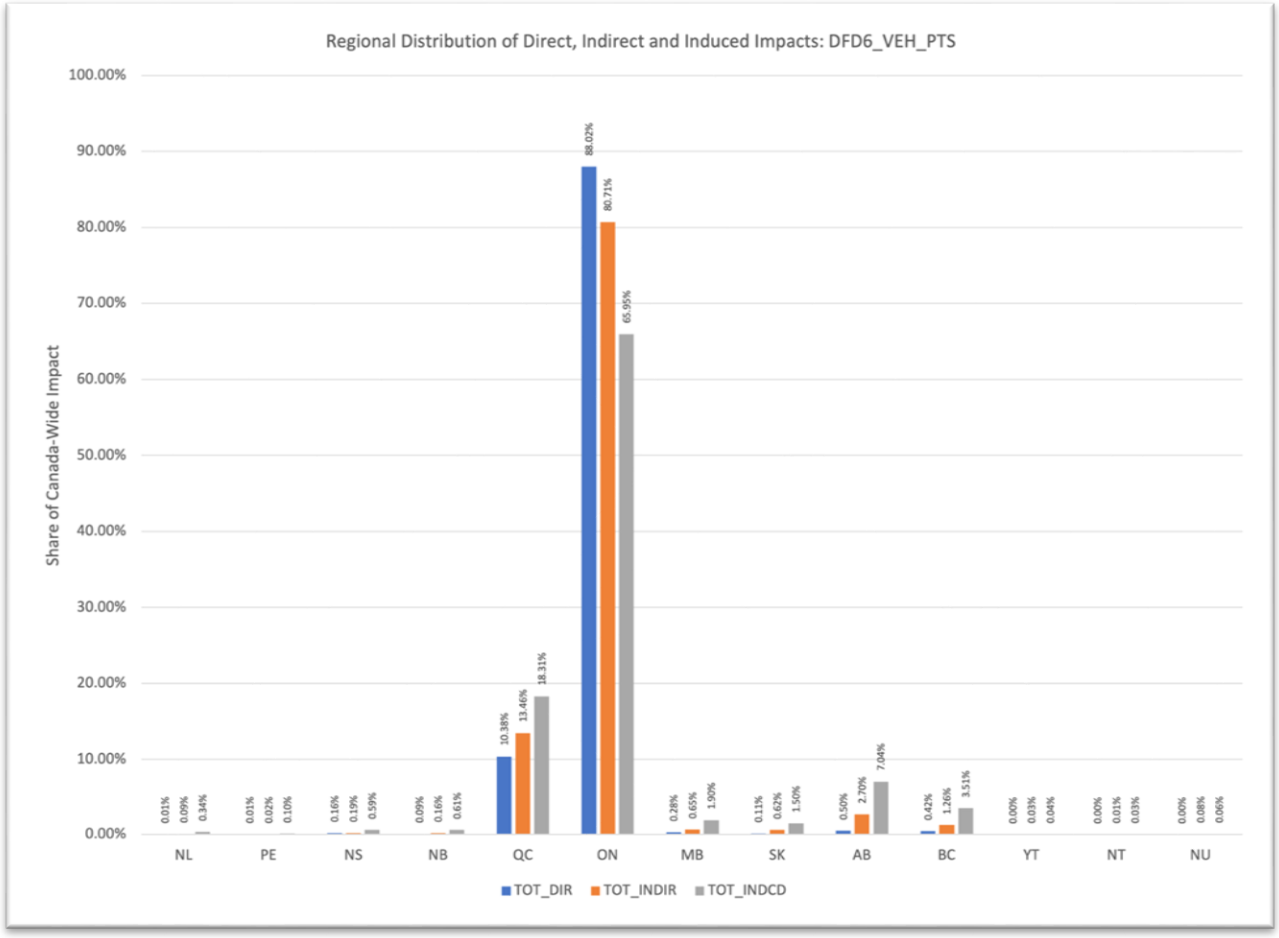


3.4 The Varying Nature of Economic Impacts Across Regions in Canada

The foregoing has made it clear that each of Canada's regions plays a somewhat unique role in its space economy – with the primary extractive emphasis of the Prairies and the secondary manufacturing emphasis of the heartland of Ontario and Quebec clearly evident. Figure 16 presents a breakdown of the total impacts associated with the most impactful scenario, DFD6_VEH_PTS, by direct, indirect and induced effects across all Canadian provinces and territories to further illustrate this result.

Figure 16 shows that DFD6_VEH_PTS effects (direct, indirect and induced) are overwhelmingly concentrated in Ontario, and to a lesser extent in Quebec, with the rest of Canada capturing a relatively smaller portion of the total impact.

Figure 16:



In contrast, Figure 17 (which presents the same information but for DFD1_HHLD_CON) shows that the economic impacts associated with a 1% reduction in household consumption expenditures stand to be distributed far more widely across Canada (though the dominance of Ontario and Quebec is still evident).

Figure 17:

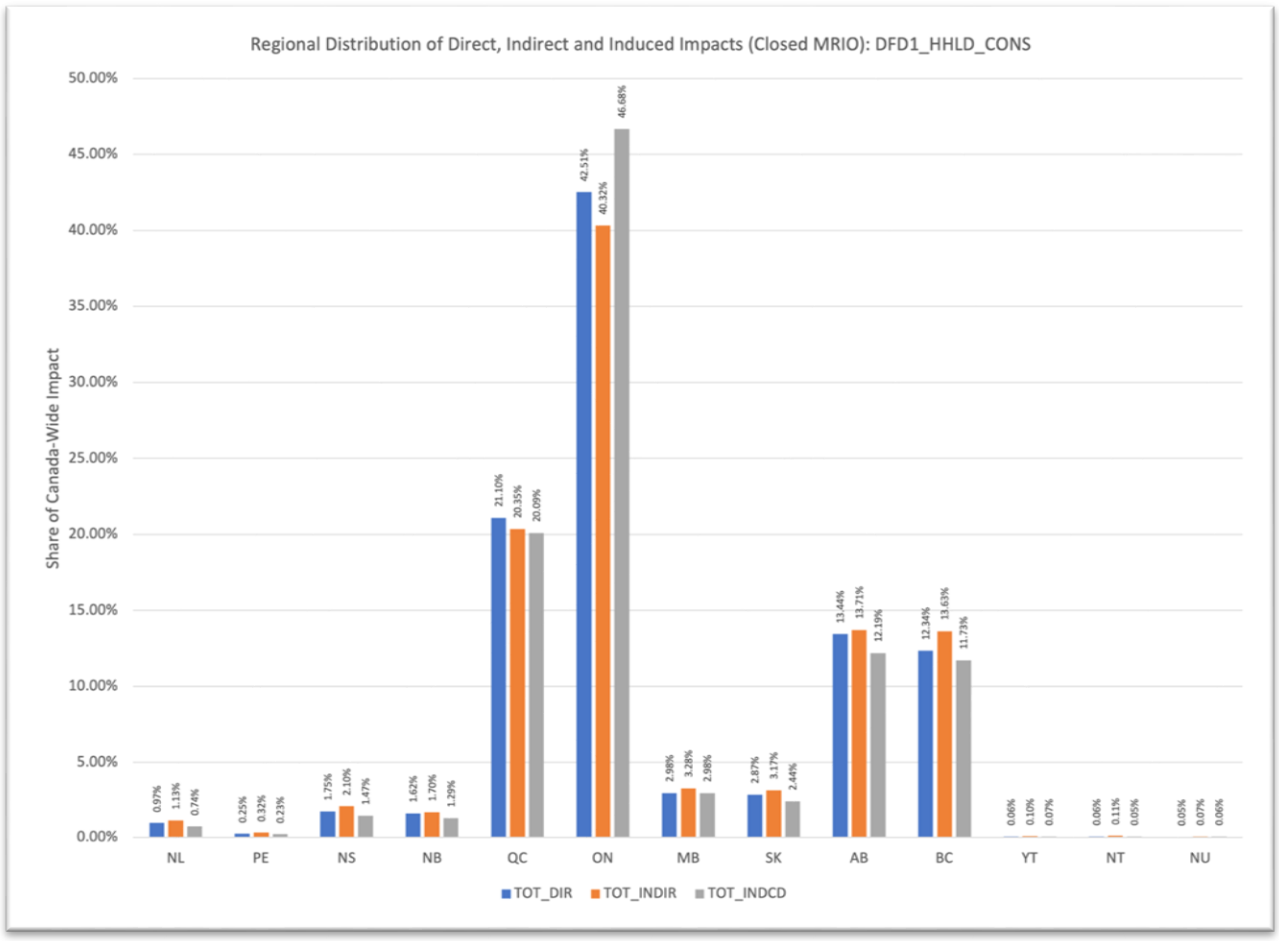

Regional Distribution of Direct, Indirect and Induced Impacts (Closed MRIO): DFD1_HHLD_CONS
Share of Canada-Wide Impact
50.00%
45.00%
40.00%
35.00%
30.00%
25.00%
20.00%
15.00%
10.00%
5.00%
0.00%
0.97% 1.13% 0.74%
0.25% 0.32% 0.23%
1.75% 2.10% 1.47%
1.62% 1.70% 1.29%
21.10% 20.35% 20.09%
42.51% 40.32% 46.68%
2.98% 3.28% 2.98%
2.87% 3.17% 2.44%
13.44% 13.71% 12.19%
12.34% 13.63% 11.73%
0.06% 0.10% 0.07%
0.06% 0.11% 0.05%
0.05% 0.07% 0.06%
NL
PE
NS
NB
QC
ON
MB
SK
AB
BC
YT
NT
NU
TOT_DIR
TOT_INDIR
TOT_INDCD

Figure 18:

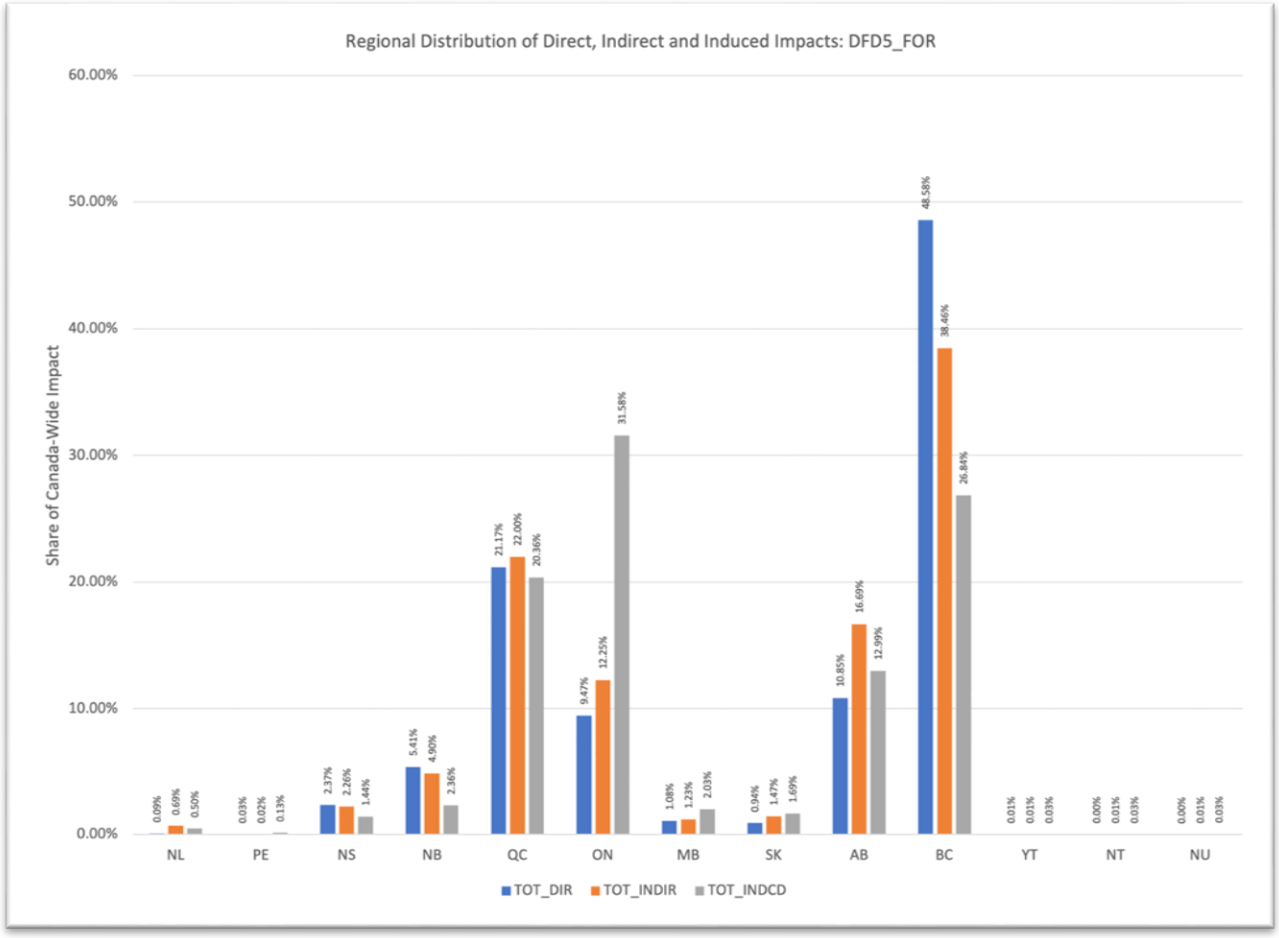
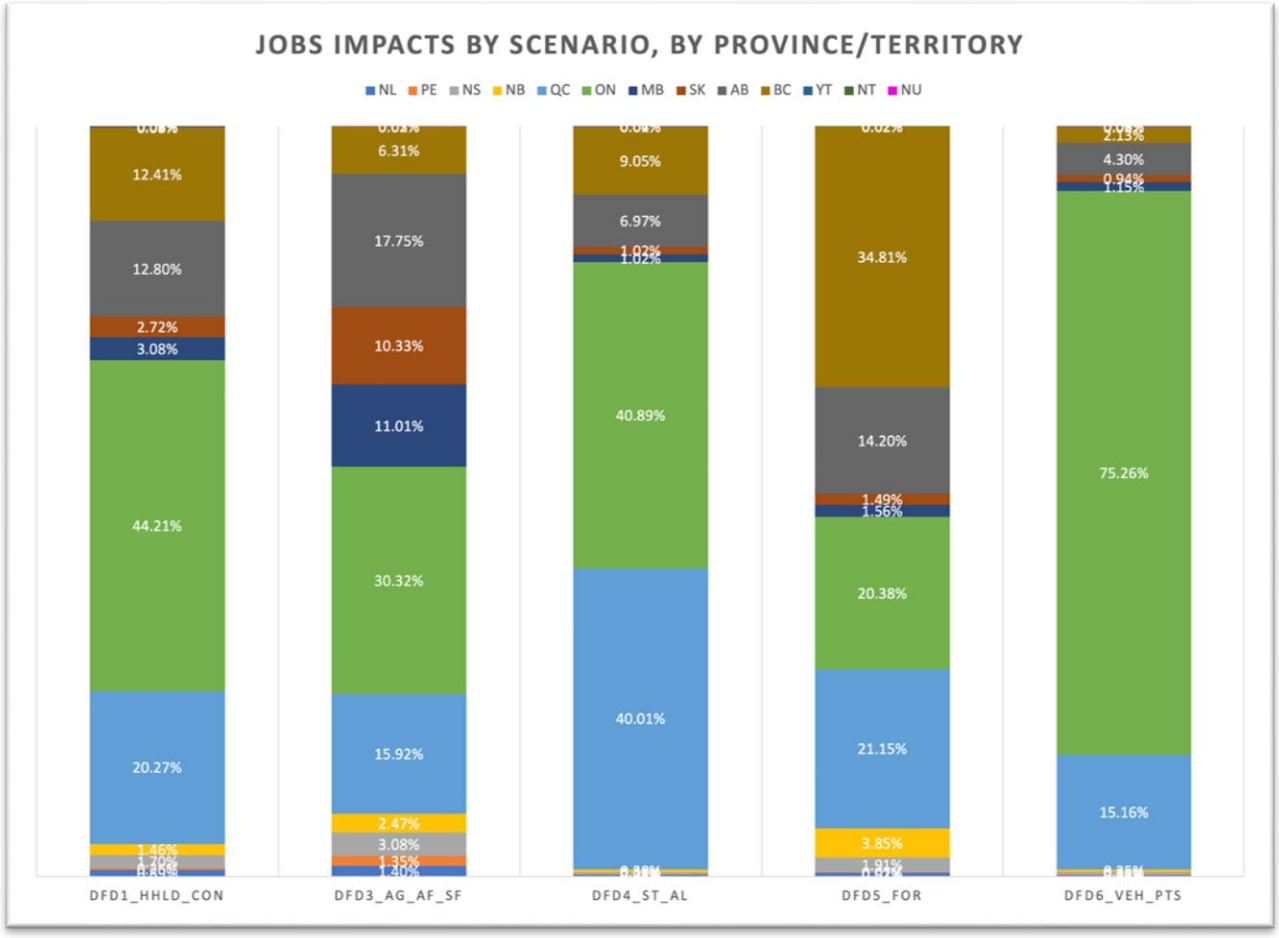


Figure 18 shows that the same is true for the effects of DFD5_FOR which posits a -12% shock to Canadian exports of softwood lumber to the U.S.. Note how the impact being captured by the Province of Ontario is largely composed of induced effects reflecting the fact that Ontario provides much of the manufactured goods involved in satisfying induced effects of the forestry and aligned sector shocks that are concentrated in the Provinces of British Columbia and Alberta (i.e., reflected by the fact that these region's capture much more pronounced direct effects).

3.4 Provincial Distribution of Employment Impacts

Figure 19 presents the regional distribution of employment impacts by scenario.

Figure 19:



As expected, Figure 19 shows that the magnitude of the employment shock and its regional distribution varies across the scenarios, with DFD6_VEH_PTS and DFD1_HHLD_CON showing the largest overall employment shocks and the highest degree of concentration in the Province of Ontario. Indeed, more than three-quarters of the total employment shock associated with the DFD6_VEH_PTS scenario lands in the Province of Ontario. Also, while the Province of Quebec is always a recipient of significant employment effects across scenarios, Quebec captures the largest share of the total employment impact under scenario DFD4_ST_AL which posits 15% and 10% reductions in exports of steel and aluminum to the U.S.. The fact that the Province of Quebec is by far Canada's largest producer of aluminum accounts for its exaggerated employment impact under this scenario relative to the others.

Figure 20:

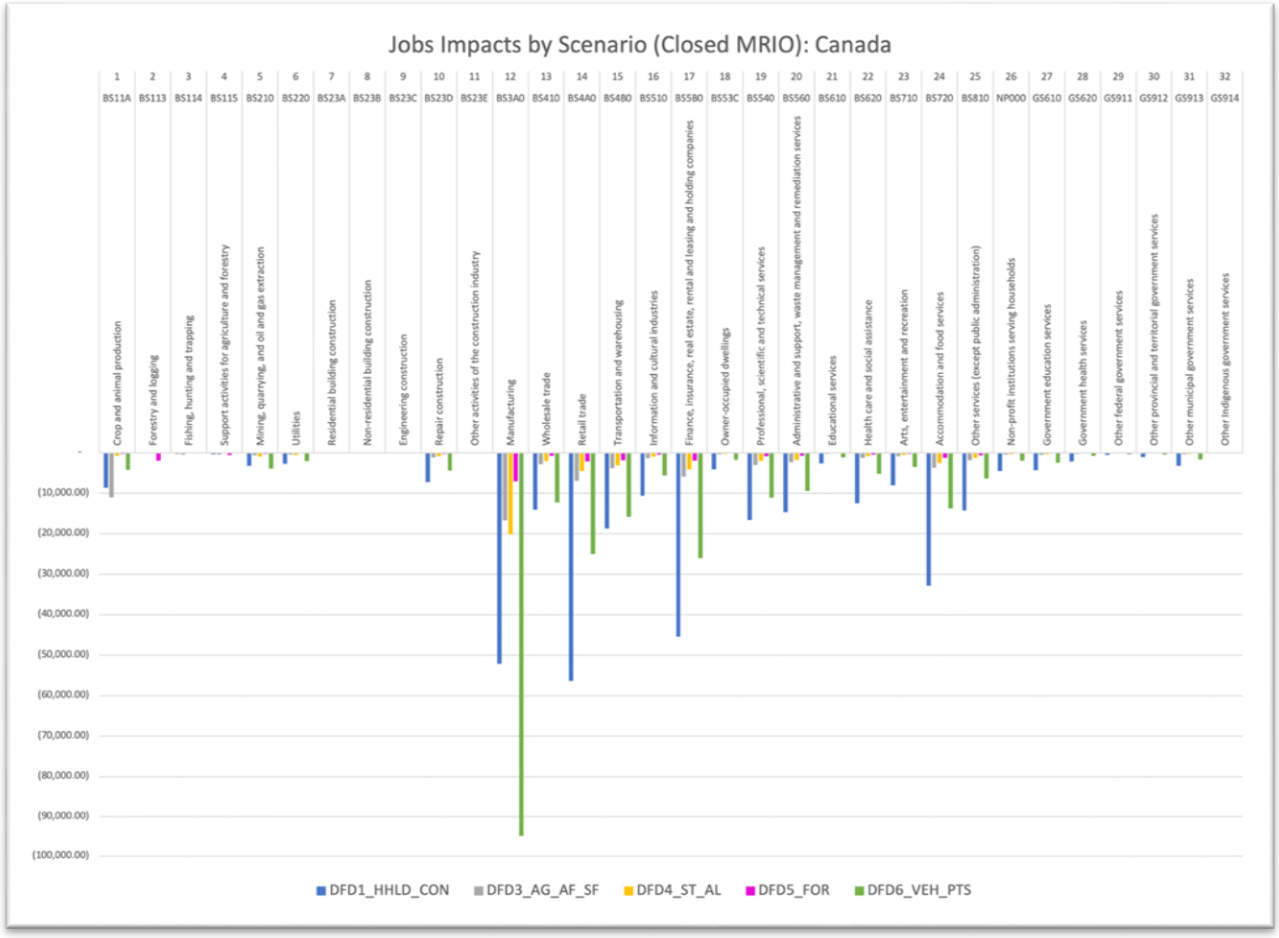


Figure 20 presents total employment impacts by industry for Canada as a whole. The manufacturing centrality of Ontario is clear to see across all scenarios, as are the secondary and tertiary induced effects which act to accentuate the regional spread.

5.0 Discussion

At the outset, a review of current thinking regarding the degree to which Canada stands to be negatively impacted by U.S. tariffs was conducted. This review led to the development of five tariff scenarios which were then analyzed using an MRIO model of the Canadian economy (using the most recent provincial/territorial IO data for Canada – 2022). The first of these scenarios focused on broad-spectrum effects relating to declining levels of household consumption expenditures in Canada. While the other four pertained to sector-specific scenarios focussed on agriculture and agri-food and seafood industries (DFD3_AG_AF_SF), steel and aluminum manufacturing (DFD4_ST_AL), softwood lumber (DFD5_FOR) and automobile and automotive parts manufacturing (DFD6_VEH_PTS).

That scenario found to generate the greatest economic impact overall using the open MRIO modek was DFD1_HHLD_CON which posits a 1% reduction in Canadian household consumption expenditures followed closely by DFD6_VEH_PTS which posits -20% and -15% shocks to exports of automobiles and automotive parts manufacturing to the US respectively. DFD_HHLD_CON is estimated to result in an impact to total gross industrial output across Canada of more than $61.4B, more than $32.2B in value-added, and the loss of 349,762 person-years of employment. DFD6_VEH_PTS generates an overall output impact which is close to that seen for DFD1_HHLD_CON ($56.25B) but far more concentrated in the Province of Ontario because of the central role played by Ontario's manufacturing core in the satisfaction of direct, indirect and induced input requirements across most industries in most regions of Canada. When the closed MRIO model is used, DFD6_VEH_PTS ranks first in terms of gross output, value-added and employment impacts.

Either of these scenarios could well represent TD Economics' (2026) "Severe Fragmentation" scenario whereby most CUSMA-related protections are repealed, and where Canada is pushed into recession. The aforementioned review also suggests that these negative impacts could manifest very quickly, certainly inside of one to two years following a termination (or significant reduction) of CUSMA.

The remaining three industry-specific scenarios give insights into how various regions and industries of Canada stand to be affected. Clearly, the manufacturing heartland of Ontario and Quebec will bear the brunt of the economic impacts associated with the cessation or significant curtailment of CUSMA. This is not to say that Canada's smaller provinces and territories will not be affected, but the severe concentration of higher-value-added manufacturing capacity in Ontario, Quebec, British Columbia, and Alberta (and particularly in the highly urbanized regions therein) suggests that the types of tariff shocks considered above could disproportionately impact these regions that are inordinately exposed to U.S. tariffs, most notably, the Province of Ontario via its singular automotive industry in the Canadian context. Other aspects of Ontario's manufacturing economy stand to be terribly impacted as well, especially in terms of the refinement of petroleum products and food processing.